COMPREHENSIVE DIGITAL PLATFORM SYSTEM REQUIREMENTS SPECIFICATION (SRS) DOCUMENTATION

For

Oromo Research Association

Version: 1.0

Document Id: SRS_001

By

NEXTEC Software Technology

November, 2025

NEX TEC Software Technology

Oromo Research Association

Project: Comprehensive Digital Platform for the Oromo Research Association (ORA)

I. Document Revision History

Version	Date	Author	Reason For Change	Description of Change
1.0	11/17/2025	Nex Tec Software Technology PLC		Initial Draft

II. Document Reviewers

Reviewer Name	Organization & Section	Position/Role	Signature	Sign-off Date

III. Document Approvers

Approver Name	Organization & Section	Position/Role	Signature	Sign-off Date

IV. Document Distribution

Name	Organization	Role	Date	Version
Oromo Research Association (ORA) Board	ORA			
Project Team	Nex Tec Software Technology PLC			

Abbreviations/Acronyms

API = Application Programming Interface

ARK = Archival Resource Key

AWS = Amazon Web Services

AWS RDS = Amazon Web Services Relational Database Service

CI/CD = Continuous Integration/Continuous Deployment

CSRF = Cross-Site Request Forgery

CSS = Cascading Style Sheets

DOI = Digital Object Identifier

DRM = Digital Rights Management

FR = Functional Requirement

GDPR = General Data Protection Regulation

HTML = HyperText Markup Language

IEEE = Institute of Electrical and Electronics Engineers

JWT = JSON Web Token

NFR = Non-Functional Requirement

ORA = Oromo Research Association

OWASP = Open Web Application Security Project

QA = Quality Assurance

RBAC = Role-Based Access Control

SOW = Statement of Work

SQL = Structured Query Language

SRS = Software Requirements Specification

UI/UX = User Interface/User Experience

UM = User Management

XSS = Cross-Site Scripting

ISSN =International Standard Serial Number

Contents

Executive Summary 1
1 Introduction 3
1.1 Purpose 3
1.2 Document Conventions 3
1.3 Intended Audience and Reading Suggestions 3
1.4 Product Scope 3
1.5 References 4
1.6 Definitions & Glossary 4
1.6.1 General System Terms 4
1.6.2 Publishing & Editorial Terms 5
1.6.3 Metadata & Identification Terms 6
1.6.4 Repository & Knowledge Management Terms 7
1.6.5 Library & Cataloging Terms 7
1.6.6 Book Processing & eBook Management Terms 8
1.6.7 System Administration & Security Terms 8
1.6.8 Notification & Communication Terms 9
1.6.9 Technical Terms 9
2 Overall Description 11
2.1 Product Perspective 11
2.2 Product Functions 11
2.3 User Classes and Characteristics 11
2.4 Operating Environment 15
2.5 Design and Implementation Constraints 15
2.6 User Documentation 16
2.7 Assumptions and Dependencies 16
3 Specific Requirements 17
3.1 External Interface Requirements 17

3.1.1 User Interfaces 17
3.1.2 Hardware Interfaces 17
3.1.3 Software Interfaces 17
3.1.4 Communications Interfaces 18
3.2 System Features (Functional Requirements) 18
3.2.1 Public Landing Page (Web-site) 18
3.2.2 Journal Management System 19
3.2.3 Online Book Publication Platform (eBooks) 20
3.2.4 Library Management System 20
3.2.5 Oromo Wikipedia Platform 21
3.2.6 Repository Management System 21
3.2.7 Researchers' Network Platform 22
3.2.8 User Management 22
3.2.9 Citation Generation Tools 23
3.3 Nonfunctional Requirements 23
3.4 Other Requirements 25
4 Analysis Models 26
4.1 List of System Use Cases 26
4.2 Use Case Diagram 27
4.3 System Use Case Detail 29
5 System Architecture & Data Requirements 87
5.1 System Architecture Overview 87
5.1.1 Architectural Style 87
5.1.2 High-Level Architecture Components 87
5.1.3 Deployment Architecture (AWS-Based) 88
5.2 System Context Diagram 88
5.3 Data Requirements 88
5.3.1 Database Requirements 88

5.3.2 Data Retention 88
5.4 Metadata Requirements Per Module 89
6 Api Requirements 90
6.1 API Style Guide 90
6.2 Authentication API Requirements 90
6.3 Versioning 90
6.4 Rate Limiting 90
7 Acceptance Criteria 91
7.1 General Acceptance Criteria 91
7.2 Module-Based Acceptance Criteria 91
8 Requirement Traceability Matrix (RTM) 94
9 Error Handling & Logging Requirements 98
9.1 Error Handling 98
9.2 Logging Requirements 98
9.3 Monitoring 99
10 Backup, Disaster Recovery & Maintenance Plan 100
10.1 Backup Requirements 100
10.2 Disaster Recovery (DR) Plan 100
10.3 Maintenance Plan 101

List of Tables

Table 1 General System Terms 4
Table 2 Publishing & Editorial Terms 5
Table 3 Metadata & Identification Terms 6
Table 4 Repository & Knowledge Management Terms 7
Table 5 Library & Cataloging Terms 7
Table 6 Book Processing & eBook Management Terms 8
Table 7 System Administration & Security Terms 8
Table 8 Notification & Communication Terms 9
Table 9 Technical Terms 9
Table 10 System-Wide Roles 12
Table 11 Journal Management System Roles 12
Table 12 eBook Publishing System Roles 13
Table 13 Library Management System Roles (Digital & Physical) 13
Table 14 Oromo Wikipedia Platform Roles 14
Table 15 ORA Repository Management System Roles 14
Table 16 Researchers' Network Platform Roles 15
Table 17 List of System Use Cases 26
Table 18 Manage System User Registration Use case 29
Table 19 Manage User Login Use case 32
Table 20 Manage Manuscript Submission Use case 35
Table 21 Manage Peer Review Submission Use case 38
Table 22 Manage Editorial Decision Use case 41
Table 23 Manage Article Publication Use case 44
Table 24 Manage Book Submission Use case 47
Table 25 Manage eBook Production Use case 50
Table 26 Search Library Catalog Use case 53
Table 27 Manage Book Lending Use case 56
Table 28 Manage Library Cataloging Use case 59
Table 29 Manage Wikipedia Article Creation Use case 62
Table 30 Manage Wikipedia Content Moderation Use case 66
Table 31 Manage Repository Deposit Use case 69
Table 32 Manage Repository Curation Use case 72
Table 33 Manage Researcher Profile Use case 75
Table 34 Manage Research Group Use case 78
Table 35 Manage Citation Generation Use case 81

Table 36 Manage Citation Export Use case 84
Table 37 Requirement Traceability Matrix (RTM) 94

Table of Figures

Figure 1 Use Case Diagram for ORA Digital Platform 28
Figure 2 Sequence Diagram for: Manage System User Registration (UC-SYS-01) 31
Figure 3 Sequence Diagram for UC-SYS-02 Manage User Login 34
Figure 4 Sequence Diagram for UC-JMS-01 Manage Manuscript Submission 37
Figure 5 Sequence Diagram for UC-JMS-04 Manage Peer Review Submission 40
Figure 6 Sequence Diagram for UC-JMS-06 Manage Editorial Decision 43
Figure 7 Sequence Diagram for UC-JMS-10 Manage Article Publication 46
Figure 8 Sequence Diagram for UC-BPS-01 Manage Book Submission 49
Figure 9 Sequence Diagram for UC-BPS-09 Manage eBook Production 52
Figure 10 Sequence Diagram for UC-LMS-01 Search Library Catalog 55
Figure 11 Sequence Diagram for UC-LMS-02 Manage Book Lending 58
Figure 12 Sequence Diagram for UC-LMS-12 Manage Library Cataloging 61
Figure 13 Sequence Diagram for UC-OWP-01 Manage Wikipedia Article Creation 65
Figure 14 Sequence Diagram for UC-OWP-03 Manage Wikipedia Content Moderation 68
Figure 15 Sequence Diagram for UC-RMS-01 Manage Repository Deposit 71
Figure 16 Sequence Diagram for UC-RMS-16 Manage Repository Curation 74
Figure 17 Sequence Diagram for UC-RNP-01 Manage Researcher Profile 77
Figure 18 Sequence Diagram for UC-RNP-02 Manage Research Group 80
Figure 19 Sequence Diagram for UC-RNP-03 Manage Citation Generation 83
Figure 20 Sequence Diagram for UC-RNP-04 Manage Citation Export 86

Executive Summary

This document provides the Software Requirements Specification (SRS) for a transformative, web-based digital platform designed to serve as the central technological ecosystem for the Oromo Research Association (ORA). This initiative addresses the critical need to centralize ORA's scholarly activities, preserve invaluable Oromo knowledge, and foster a connected global community of researchers. The proposed platform will replace fragmented processes with a single, integrated, and modern solution.

The proposed solution is a comprehensive, cloud-hosted platform composed of six seamlessly integrated modules: a Journal Management System, an Online Book Publication (eBook) Platform, a Library Management System, a community-driven Oromo Wikipedia, a secure Repository Management System, and a collaborative Researchers' Network Platform. Together, these components will create a dynamic digital hub that supports the entire lifecycle of research from creation and peer review to publication, archival, and community engagement.

The primary objective of this project is to empower the Oromo Research Association to achieve its mission of promoting Oromo culture, language, and academic excellence through technology. The key strategic benefits include:

- Streamlining Academic Publishing: Automating the submission, peer review, and publication workflows for journals and books.
- Enhancing Knowledge Preservation: Providing a secure, permanent digital archive for research papers, historical documents, and datasets.
- Fostering Global Collaboration: Creating a dedicated networking space for Oromo researchers and academics worldwide to connect, share ideas, and collaborate on projects.
- Expanding Global Outreach: Making Oromo-centered knowledge accessible to a global audience of students, educators, and the public through open-access repositories and a dedicated Wikipedia.

The platform will be developed using a modern, scalable, and secure technology stack, featuring React.js for a responsive user interface, Node.js for a robust backend, and MySQL for reliable data management. Adherence to industry best practices, including the OWASP Top 10 security standards and deployment on a reliable cloud infrastructure like AWS, will ensure the platform is robust, secure, and highly available. The system is designed with a modular architecture to be maintainable and extensible, allowing for future growth and the addition of new features.

The project is planned over a six-month duration, structured into three distinct phases:

1. Month 1: Analysis & Design (Requirement finalization, architecture design, and UI/UX mockups).

2. Months 2-5: Backend & Frontend Development (Building and integrating all six modules).
3. Month 6: Quality Assurance, Deployment & Training (Comprehensive testing, security auditing, deployment to a live server, and handover with full documentation and training for ORA staff).

In conclusion, this project represents a significant strategic investment in the digital future of the Oromo Research Association. The successful delivery of this platform will not only elevate ORA's operational capabilities but will also create a lasting and accessible digital legacy, solidifying its role as a premier center for Oromo scholarship and research for years to come.

1 Introduction

1.1 Purpose

The purpose of this document is to provide a detailed description of the requirements for the Oromo Research Association (ORA) Digital Platform. It will outline the system's purpose, features, and operational constraints, serving as a foundational agreement between ORA and Nex Tec Software Technology PLC for development.

1.2 Document Conventions

This document is based on the IEEE 830-1998 standard for SRS documents. All requirements are uniquely identifiable. The following keywords are used:

- **Shall:** Indicates a mandatory requirement that must be implemented.
- **Should:** Indicates a recommended feature that is desirable but not mandatory.
- **Will:** Describes a fact or a declaration of purpose.

1.3 Intended Audience and Reading Suggestions

This SRS is primarily intended for the following audiences:

- **ORA Stakeholders:** Review the **Introduction** and **Overall Description** sections for a high-level overview and to confirm project scope and objectives.
- **Project Managers:** Use this document for project planning, tracking, and scope management.
- **Developers & Architects:** Focus on the **Operating Environment**, **Design and Implementation Constraints**, and **Specific Requirements** sections for detailed technical constraints and functional details.
- **QA & Testing Teams:** Use the **Specific Requirements** section to create test cases and verify that the implementation meets all specified requirements.
- **Documentation Writers:** Responsible for creating user manuals, training materials, and other documentation for the system.

1.4 Product Scope

This project will deliver a complete web-based digital platform for ORA. The platform will be a cloud-hosted, integrated ecosystem combining six essential modules:

1. Journal Management System

2. Online Book Publication Platform (eBooks)
3. Library Management System
4. Oromo Wikipedia Platform
5. Repository Management System
6. Researchers' Network Platform
7. Landing Page (Web-site)

The scope includes full-stack development, integration of all modules, security auditing, deployment, training, and handover.

The following items are explicitly **out of scope** for this project:

- Development of native mobile applications (iOS/Android). The platform will be mobile-responsive.
- Content creation, data entry, and migration of existing physical documents.
- Ongoing hosting, domain, and maintenance costs post-handover.
- Procurement of any hardware.

1.5 References

- Oromo Research Association (ORA) - Terms of Reference
- Nex Tec Software Technology PLC - Technical Proposal for ORA Digital Platform

1.6 Definitions & Glossary

This section provides definitions of terms, concepts, and domain-specific vocabulary used throughout the ORA Digital Publishing & Knowledge Management System. These definitions ensure common understanding among stakeholders, developers, reviewers, and end users.

1.6.1 General System Terms

Table 1 General System Terms

Term	Definition
System	The complete ORA Digital Platform, including Journal Management System, Book Processing System, eBook Management, Library Management System, Repository, Encyclopedia/Wiki, and Citation Tool.

Module	A functional component of the system that performs a distinct set of tasks such as Journal Management, Library, or Repository.
User Role	A defined profile assigned to system users (e.g., Author, Editor, Reviewer) that determines what actions they are allowed to perform.
Dashboard	A personalized summary interface displaying metrics, notifications, shortcuts, and assigned tasks based on the user's role.
Workflow	A structured sequence of automated or human actions performed to complete a process such as manuscript review or book publication.
Record	A stored data item within the system (e.g., article, book, user profile, repository item).

1.6.2 Publishing & Editorial Terms

Table 2 Publishing & Editorial Terms

Term	Definition
Manuscript	A scholarly paper submitted by an author for consideration in a journal. It typically includes the main document and supplementary files.
Submission	The process by which an author uploads a manuscript and fills in metadata required for the review process.
Peer Review	A formal evaluation of a manuscript by independent subject experts to determine originality, quality, and suitability for publication.
Double-Blind Review	A review model where both author and reviewer identities are hidden from each other.
Reviewer Assignment	The process by which an Editor or Section Editor selects qualified reviewers for a manuscript.
Editor-in-Chief (EIC)	The highest-level editorial manager responsible for final publication decisions.
Section Editor	An editor responsible for managing manuscripts within a specific discipline or journal section.
Editorial Decision	A formal decision such as "Accept", "Minor Revisions", "Major Revisions", or "Reject".

Revision Cycle	The iterative process where authors revise their manuscripts in response to reviewer comments.
Camera-Ready Version	The final, publication-ready version of an article after all corrections are applied.
Galley Proof	A pre-publication formatted version of an article sent to authors for proofreading.
Publication Issue	A collection of articles grouped by volume and number (e.g., Vol. 12, Issue 2).
Retraction	Removal of a published article due to major issues such as plagiarism or scientific misconduct.

1.6.3 Metadata & Identification Terms

Table 3 Metadata & Identification Terms

Term	Definition
Metadata	Structured descriptive information about a document, dataset, article, or book (e.g., title, author, keywords).
Dublin Core	An international metadata standard used for describing digital resources (e.g., title, creator, subject).
DOI (Digital Object Identifier)	A unique permanent identifier assigned to digital objects such as journal articles.
ISBN (International Standard Book Number)	A unique identifier assigned to books and eBooks.
ISSN (International Standard Serial Number)	A unique code assigned to journals and serial publications.
Handle/ARK	Persistent identifiers used in repository systems for long-term access to deposited materials.
ORCID	A persistent digital identifier used to distinguish researchers and link their scholarly activities.

1.6.4 Repository & Knowledge Management Terms

Table 4 Repository & Knowledge Management Terms

Term	Definition
Repository Deposit	Submission of research outputs (papers, datasets, theses, reports) into the institutional repository.
Institutional Repository (IR)	A digital archive where the organization stores and preserves scholarly assets for long-term access.
Embargo Period	A delay period during which deposited items are restricted from public access.
Preservation Copy	A high-quality version of a file stored specifically for long-term archiving.
Access Rights	Permissions that determine who can view, download, or reuse a deposited item.
Digital Asset	Any digital file stored in the system, including PDFs, images, eBooks, audio, and datasets.

1.6.5 Library & Cataloging Terms

Table 5 Library & Cataloging Terms

Term	Definition
OPAC (Online Public Access Catalog)	The user-facing interface for searching and browsing the library's collection.
Cataloging	The process of assigning metadata, classification numbers, and identifiers to physical or digital resources.
Classification Number	A numeric code (e.g., Dewey Decimal Classification) used to categorize books on shelves.
Circulation	Core library function covering lending, returning, renewing, and reserving physical books.
Hold/Reservation	A request placed by a user to borrow a book currently checked out.

Acquisition	The process of selecting, ordering, and purchasing new materials for the library.
Weeding	Removal or deactivation of outdated or damaged materials from the library collection.

1.6.6 Book Processing & eBook Management Terms

Table 6 Book Processing & eBook Management Terms

Term	Definition
Book Proposal	A formal submission by authors/editors requesting approval for a new book project.
Manuscript Package	A collection of all book chapters and related materials uploaded by multiple contributors.
Copyediting	The editorial process of correcting grammar, formatting, and consistency in the manuscript.
Typesetting	Converting the manuscript into a formatted layout for print or digital publication.
ePub Format	A widely supported eBook file format.
PDF eBook	A fixed-layout digital book format.
Book Production Workflow	Complete life cycle from proposal, review, editing, layout, to final publication.
Royalty Report	A financial statement showing earnings from sales or usage of a book/eBook.

1.6.7 System Administration & Security Terms

Table 7 System Administration & Security Terms

Term	Definition
RBAC (Role-Based Access Control)	Authorization model where permissions depend on assigned roles.
Super Admin	The highest-level administrator with unrestricted system access.
Authentication	Process of verifying the identity of a user (login, MFA).

Authorization	Process that determines what a verified user can access or perform.
Session Timeout	A security measure that logs out inactive users automatically.
Audit Trail	A historical log of user actions for accountability and compliance.
Encryption	The use of cryptographic methods to protect data at rest and in transit.
Backup & Recovery	The process of saving and restoring system data for business continuity.

1.6.8 Notification & Communication Terms

Table 8 Notification & Communication Terms

Term	Definition
System Notification	Automated messages informing users about tasks, deadlines, decisions, or system events.
Email Template	Predefined structured email used for system-generated communications.
In-App Notification	Message displayed within the system interface.
Broadcast Message	A message sent by administrators to all users or selected groups.

1.6.9 Technical Terms

Table 9 Technical Terms

Term	Definition
API (Application Programming Interface)	A set of endpoints that allow system modules or external systems to communicate.
REST API	A style of API using HTTP requests and JSON responses.
Frontend	The user interface built with web technologies.
Backend	Server-side application logic and APIs.
Cloud Storage	Remote storage infrastructure (e.g., S3) used for saving system assets.

CDN (Content Delivery Network)	A network used to deliver content quickly to global users.
Indexing	The process of creating optimized search structures for fast retrieval.

2 Overall Description

2.1 Product Perspective

The ORA Digital Platform is a new, self-contained system that will serve as the central technological hub for ORA's activities. It will operate as a dynamic digital ecosystem designed to centralize scholarly content, preserve cultural knowledge, and foster a global research community, replacing fragmented or manual processes currently in use.

2.2 Product Functions

The system will provide the following core functions:

- **Academic Publishing:** Facilitate submission, peer review, and publication of academic journals.
- **Digital Distribution:** Enable the publication and distribution (free or monetized) of eBooks.
- **Collection Management:** Catalog and manage both physical and digital library collections.
- **Knowledge Sharing:** Provide a collaborative, community-based platform for encyclopedic content (Oromo Wikipedia).
- **Digital Archiving:** Offer a secure repository for research papers, datasets, and theses.
- **Academic Networking:** Create a professional space for researchers to connect, collaborate, and share work.

2.3 User Classes and Characteristics

The ORA Digital Platform will support a variety of user roles, each with specific permissions and responsibilities tailored to their function within the ecosystem. These roles are categorized below by their primary system module.

A. System-Wide Roles

User Class	Description	Key Responsibilities
System Administrator	A technical user with the highest level of permissions, responsible for the overall health, security, and configuration of the platform.	Manages user accounts and access rights across all modules, performs system backups, ensures platform security, handles technical troubleshooting, and manages backend infrastructure.

Public User / Reader	Any non-authenticated visitor to the platform. This represents the largest group of users.	Searches for and discovers public content (articles, books, etc.), reads/views open-access materials, and can cite content using the platform's tools.

Table 10 System-Wide Roles

B. Journal Management System Roles

User Class	Description	Key Responsibilities
Journal Manager	An administrative role focused on the operational aspects of a specific journal.	Configures journal settings, sections, and submission policies; maintains the peer-review workflow; and oversees the technical performance of their journal.
Editor-in-Chief (EIC)	The senior academic leader with ultimate authority over a journal's content and quality.	Makes the final accept/reject decision on all manuscripts, assigns Associate Editors, ensures academic and ethical compliance, and oversees the entire publication process.
Associate Editor	A subject-matter expert who manages the peer-review process for assigned manuscripts. Equivalent to a "Handling Editor."	Conducts initial manuscript screening (desk rejects), selects and invites appropriate reviewers, evaluates reviewer feedback, and makes a recommendation to the EIC.
Reviewer / Referee	An external subject-matter expert who provides a critical evaluation of a manuscript.	Reviews manuscript content for methodology, ethics, and quality; provides structured, confidential feedback; and adheres to blinded review policies.
Author	A researcher submitting their work for publication.	Submits manuscripts with all required metadata, responds to reviewer comments by uploading revisions, and ensures originality and ethical compliance.

Table 11 Journal Management System Roles

C. eBook Publishing System Roles

User Class	Description	Key Responsibilities
Author / Researcher	The individual submitting a manuscript for book publication.	Prepares and submits manuscripts with metadata, responds to peer-review feedback, and approves the final proof.

Book Editor	A designated ORA staff member who oversees the entire book publishing workflow from submission to acceptance.	Performs initial manuscript screening, assigns peer reviewers, makes editorial decisions (accept, revise, reject), and communicates with authors.
Digital Content Manager	A technical production role responsible for creating the final eBook product.	Validates file quality, converts manuscripts to PDF/EPUB formats, uploads final eBooks, assigns metadata (ISBN, DOI), and sets access permissions.
Finance & Operations Officer	An administrative role managing the financial aspects of book publication.	Manages payments for Book Processing Charges (BPC), validates payments, issues invoices/receipts, and approves/declines fee waiver requests.

Table 12 eBook Publishing System Roles

D. Library Management System Roles (Digital & Physical)

User Class	Description	Key Responsibilities
Library Manager	Oversees all library operations, staff, and policies for both the digital and physical collections.	Sets circulation policies, supervises staff (Librarians, Catalogers, etc.), approves acquisitions, and generates usage reports for planning.
Digital Librarian	Manages the digital resources within the library system.	Uploads and manages digital content, ensures metadata accuracy, organizes digital collections, and monitors usage statistics.
Librarian (Physical)	Manages the day-to-day services for the physical library.	Manages lending and returning of items, handles holds and renewals, collects fines, and assists patrons directly.
Cataloger	A specialist who classifies and catalogs physical materials for easy retrieval.	Classifies items using DDC/LCC standards, assigns call numbers and barcodes, and maintains the accuracy of the physical catalog database.
Inventory Manager	Maintains the physical inventory and performs audits.	Conducts stocktaking and regular audits, manages item tagging (barcode/RFID), and tracks inventory accuracy.
Member / User / Student	The end-user of the library services.	Searches the catalog, borrows and returns items (physical and digital), places holds/reservations, and views their own borrowing history.

Table 13 Library Management System Roles (Digital & Physical)

E. Oromo Wikipedia Platform Roles

User Class	Description	Key Responsibilities
Registered Editor	A community member who actively creates and edits articles in Afaan Oromo.	Creates new articles, edits existing content, uploads free-license images/media, and participates in policy discussions.
Administrator (Sysop)	A trusted user with elevated rights to manage content and user behavior to maintain the wiki's integrity.	Deletes/restores pages, blocks vandals and disruptive IPs, protects sensitive pages from edits, and closes deletion discussions.
Bureaucrat & Global Steward	A senior user responsible for managing user roles and global platform actions.	Promotes or demotes local administrators, renames user accounts globally, and oversees governance policies.
Oversighter / CheckUser	A highly trusted user with access to sensitive information for abuse control, bound by strict privacy guidelines.	Suppresses revisions containing private data (e.g., GDPR compliance) and views user IP addresses only in serious cases of abuse.

Table 14 Oromo Wikipedia Platform Roles

F. ORA Repository Management System Roles

User Class	Description	Key Responsibilities
Researcher / Author	The individual depositing their scholarly work into the repository.	Uploads documents and datasets, provides complete and accurate bibliographic metadata (Dublin Core), and specifies the access level (Open/Restricted).
Repository Curator	A trusted staff member who manages deposits, metadata, and access policies.	Validates metadata quality, enriches records with controlled vocabularies, verifies copyright policies, and applies access controls.
Content Reviewer	A subject-matter expert who verifies the academic integrity of submissions.	Assesses the academic quality and relevance of a submission, checks for plagiarism, and recommends approval or revision.
Repository Administrator	Oversees the overall operations and policies of the repository.	Makes the final approval on all submissions, manages access control policies, and generates analytics reports.

Table 15 ORA Repository Management System Roles

G. Researchers' Network Platform Roles

User Class	Description	Key Responsibilities
Researcher / Member	An academic who uses the platform to connect, collaborate, and share work.	Creates and maintains a detailed professional profile, searches for and connects with peers, and participates in messaging, forums, and groups.
Group Moderator	A researcher who oversees a specific research group.	Approves and manages group memberships, moderates group discussions, and ensures adherence to community guidelines.
Event / Content Manager	A curation role focused on keeping the community engaged and informed.	Publishes announcements for journal calls, conferences, and events; keeps platform content up-to-date; and sends notifications to users.
Platform Administrator	Manages the settings, user accounts, and policies of the networking platform.	Manages user registrations and roles, oversees security and privacy settings, and handles system maintenance and updates.

Table 16 Researchers' Network Platform Roles

2.4 Operating Environment

The system will be deployed in a cloud-based environment (e.g., AWS) with the following architecture:

- **Frontend:** Hosted via S3 or a Nginx server.
- **Backend:** Services running on EC2 instances or in Docker containers.
- **Database:** Provisioned with RDS (AWS) or MySQL on a managed instance.
- **Deployment:** CI/CD pipeline using GitHub Actions or Jenkins.

2.5 Design and Implementation Constraints

- **Frontend:** React.js (with Ant Design and Tailwind CSS).
- **Backend:** Node.js (Express) or Spring Boot (Java).
- **Database:** PostgreSQL
- **Architecture:** Multi-tier (Presentation, Application, Data, Security layers).
- **Security Protocols:** Secure API access using JWT and HTTPS.
- **Standards:** Must comply with metadata standards (e.g., Dublin Core) and citation formats (APA, MLA).

2.6 User Documentation

Comprehensive user and administrator documentation will be provided as part of the final handover. This will include guides for content managers and system administrators.

2.7 Assumptions and Dependencies

- ORA will provide a dedicated point of contact for timely feedback and decision-making throughout the project lifecycle.
- ORA is responsible for ensuring all content published on the platform complies with copyright and data privacy regulations (e.g., GDPR).
- The system's performance is dependent on the selected cloud hosting plan and the stability of third-party services (e.g., AWS RDS).

3 Specific Requirements

3.1 External Interface Requirements

This section specifies the external interfaces of the ORA Digital Platform, including how it will interact with users, other software systems, and hardware.

3.1.1 User Interfaces

The platform shall provide a single, consistent, and user-friendly graphical user interface (GUI) for all modules.

- **FR-EI-001 (Responsiveness):** The user interface shall be fully responsive and provide an optimal user experience on a wide range of devices, including desktops, tablets, and mobile smartphones.
- **FR-EI-002 (Accessibility):** The UI shall adhere to the **Web Content Accessibility Guidelines (WCAG) 2.1 Level AA** standards to ensure it is accessible to users with disabilities, including those who use screen readers.
- **FR-EI-003 (Browser Compatibility):** The system shall be fully functional on the latest two versions of all major web browsers, including Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.

3.1.2 Hardware Interfaces

There are no direct, specific hardware interfaces required for the end-users beyond standard client devices (computers, tablets). However, the system shall support the following peripherals for administrative roles:

- **FR-EI-004 (Barcode Scanners):** The Library Management System's circulation module shall support standard USB plug-and-play barcode scanners for the efficient check-in and check-out of physical books.

3.1.3 Software Interfaces

The ORA Digital Platform will interface with several external software systems via their Application Programming Interfaces (APIs) to provide its full range of functionality.

- **FR-EI-005 (DOI Registration Agency):** The system shall securely integrate with the **Crossref API**. This interface will be used by the Journal Management and eBook Publishing modules to programmatically register publications and receive official DOIs.
- **FR-EI-006 (Payment Gateway):** The eBook Publishing module shall integrate with a designated payment gateway (e.g., **Stripe, PayPal, or a local provider like Chapa**) to

securely process payments for monetized eBooks. This interface must handle transaction processing and confirmation.

- **FR-EI-007 (Email Service):** The system shall integrate with a reliable transactional email service (e.g., **Amazon SES or SendGrid**) to handle the sending of all automated system emails, including user account verifications, peer review notifications, and password resets.
- **FR-EI-008 (Bibliographic Databases - Optional):** The Library Management System's cataloging module should support metadata import from external library databases (e.g., Library of Congress) using the **Z39.50 protocol** to facilitate rapid cataloging of new books via their ISBN.

3.1.4 Communications Interfaces

This section defines the communication protocols that will be used for data transfer between the system and its users, as well as between internal components.

- **FR-EI-009 (Primary Protocol - HTTPS):** All communication between the end-user's web browser and the system's web server shall be encrypted using **HTTPS (Hypertext Transfer Protocol Secure)** with valid SSL/TLS certificates to ensure data confidentiality and integrity.
- **FR-EI-010 (API Communication - REST):** Communication between the frontend client (React.js) and the backend server (Node.js) shall be handled via a secure **RESTful API** using the JSON data format.
- **FR-EI-011 (Database Communication):** The backend application server shall communicate with the PostgreSQL database using the standard, secure database connection protocol provided by PostgreSQL.
- **FR-EI-012 (Email Communication - SMTP):** The system shall use the **Simple Mail Transfer Protocol (SMTP)** to send outgoing emails via the integrated external email service.

3.2 System Features (Functional Requirements)

3.2.1 Public Landing Page (Web-site)

This module represents the main public-facing website and entry point for the entire ORA Digital Platform. It is designed to engage visitors, communicate ORA's mission, and provide intuitive pathways to the platform's rich content.

- **FR-LP-001: (Brand Identity and Mission Statement)** The landing page shall prominently display the Oromo Research Association's official logo and a clear, concise mission statement that communicates the purpose and vision of ORA.

- **FR-LP-002: (Primary Navigation)** The page must feature a clear and persistent main navigation menu (header) that provides direct links to all the primary modules of the platform: Journals, Books, Library, Oromo Wikipedia, Repository, and the Researchers' Network. It must also include "Login" and "Register" buttons.
- **FR-LP-003: (Dynamic Content Showcase)** The landing page shall not be static. It must feature dynamic sections that automatically update to showcase new and noteworthy content. This shall include sections for:
    - Recently Published Journal Articles
    - New eBook Arrivals
    - Featured Oromo Wikipedia Entries
    - ORA News and Announcements (This content shall be managed by the **Event/Content Manager** or **System Administrator**)
- **FR-LP-004: (Global Search Bar)** A prominent and easily accessible search bar must be featured on the landing page, allowing users to perform a global search across all public content on the platform.
- **FR-LP-005: (Calls to Action - CTAs)** The page shall include clear calls to action to guide user engagement, such as "Submit Your Manuscript," "Join Our Network," or "Explore the Repository."
- **FR-LP-006: (Standard Footer)** A comprehensive footer section must be present on the page, containing links to essential information such as "About ORA," "Contact Us," "Terms of Service," and "Privacy Policy."

3.2.2 Journal Management System

- **FR-JM-001:** An **Author** shall be able to submit a manuscript in various formats (e.g., .docx, .pdf).
- **FR-JM-002:** The system shall allow an **Associate Editor** to assign submitted manuscripts to reviewers for anonymous (double-blind) peer review.
- **FR-JM-003: A Reviewer** shall be able to submit their comments and a recommendation (Accept, Minor Revisions, Major Revisions, Reject) via the platform.
- **FR-JM-004:** The system shall provide a dashboard for **Editor-in-Chief** and **Associate Editor** to track the status of all manuscripts in the review pipeline.
- **FR-JM-005:** Upon acceptance, the **Journal Manager** shall facilitate the publication of the article into a specific journal volume/issue.

- **FR-JM-006:** (ISSN Management) The system shall allow a **Journal Manager** to input and store the unique ISSN for each journal series managed by the platform. This ISSN shall be prominently displayed on the journal's main homepage and included in the metadata of every article published within that journal.
- **FR-JM-007:** (DOI Generation and Assignment) The system shall allow a **Journal Manager** to initiate the automatic generation and assignment of a unique **DOI** to each individual article upon its official publication. This will be achieved by integrating with a recognized DOI registration agency (e.g., Crossref).
- **FR-JM-008:** (DOI Display and Resolution) The assigned DOI shall be stored permanently with the article's metadata. It must be displayed as a permanent, clickable link on the article's abstract page and included in all citation formats (e.g., APA, MLA) generated by the system's citation tools.

3.2.3 Online Book Publication Platform (eBooks)

- **FR-BP-001: A Digital Content Manager** shall be able to upload eBook files (EPUB, PDF) and associate them with metadata (**ISBN, DOI, author, keywords, etc.**).
- **FR-BP-002:** The system shall provide an in-browser reader for both EPUB and PDF formats with features like bookmarking and table of contents navigation.
- **FR-BP-003:** The system shall support monetized access, including allowing a **Finance & Operations Officer** to manage payments through an integrated payment gateway.
- **FR-BP-004:** The platform shall implement DRM protection to prevent unauthorized copying and distribution of purchased content.

3.2.4 Library Management System

- **FR-LM-001:** A **Digital Librarian** (for digital content) and a **Cataloger** (for physical content) shall be able to catalog both physical (hard copy) and digital (soft copy) books in the system.
- **FR-LM-002:** Each catalog entry for a physical book shall include fields for title, author, subject, year, ISBN, and physical location/shelf number.
- **FR-LM-003:** The system shall provide an advanced search feature for all users to find books by filtering on title, author, subject, and publication year.
- **FR-LM-004:** The system shall display the real-time availability status for physical books (e.g., "Available," "Checked Out," "On Hold").
- **FR-LM-005:** A **Member / User / Student** shall be able to place a hold or reservation on a physical book that is currently checked out.

- **FR-LM-006:** The system shall support digital borrowing for soft-copy materials, allowing a user to access a digital book for a predefined loan period.
- **FR-LM-007:** The platform shall include a dedicated dashboard for **Librarian (Physical)** and **Library Manager, and Inventory Manager** to manage inventory, track check-ins/check-outs, and oversee reservations.
- **FR-LM-008:** The system should support barcode scanning for efficient management of the physical book inventory.

3.2.5 Oromo Wikipedia Platform

- **FR-OW-001:** A **Registered Editor** shall be able to create new articles using a Wikipedia-style rich-text editor.
- **FR-OW-002:** A **Registered Editor** shall be able to edit existing articles on the platform.
- **FR-OW-003:** The system shall maintain a complete version history for every article, allowing users to view past versions and see changes between revisions.
- **FR-OW-004:** The platform shall support content creation and display in multiple languages, specifically Afaan Oromo and English.
- **FR-OW-005:** The system shall implement a role-based moderation workflow where an **Administrator (Sysop)** must approve new articles or significant edits before they become publicly visible.
- **FR-OW-006: Registered Editors** shall be able to add tags and assign categories to articles to improve organization and discoverability.
- **FR-OW-007:** Each article shall have an associated "talk" or "discussion" page for users to collaborate and discuss the content of the article.

3.2.6 Repository Management System

- **FR-RM-001:** A **Researcher / Author** shall be able to upload various types of scholarly documents, including research papers, theses, dissertations, conference papers, and datasets.
- **FR-RM-002:** The system shall require the uploader to provide metadata for each submission based on the Dublin Core standard (or a similar standard), including fields for creator, title, date, and identifier.
- **FR-RM-003:** The uploader shall be able to set the access level for each submitted item as either "Open Access" (publicly available) or "Restricted Access" (available only to specific user roles or upon request).

- **FR-RM-004:** The system shall automatically generate a persistent and unique URL (e.g., a Handle or ARK) for every item in the repository to ensure stable linking over time.
- **FR-RM-005:** All users shall be able to search and filter the repository's contents using keywords, author, date, and document type.
- **FR-RM-006:** The system shall provide functionality for **Repository Administrator** or **System Administrator** to perform regular backups of the repository's contents for data protection and disaster recovery.
- **FR-RM-007: (DOI for Repository Items)** The system should have the capability to assign DOIs to specific items in the repository, such as significant datasets, theses, or original reports. This ensures that these valuable scholarly assets are permanently findable and easily citable by other researchers.

3.2.7 Researchers' Network Platform

- **FR-RN-001:** A **Researcher / Member** shall be able to create and customize a personal profile, including fields for academic credentials, affiliations, publications, and research interests.
- **FR-RN-002:** A **Researcher / Member** shall be able to search the network of researchers by name, field of study, or institution.
- **FR-RN-003:** The system shall provide a private, internal messaging system for researchers to communicate one-on-one.
- **FR-RN-004:** The platform shall support the creation of public or private discussion forums and research groups focused on specific topics.
- **FR-RN-005:** The system shall provide tools for collaboration within a research group, such as a shared document space and a group-specific discussion board.
- **FR-RN-006:** Users shall receive notifications within the platform for relevant activities, such as new messages, connection requests, or updates in their research groups.

3.2.8 User Management

- **FR-UM-001:** The system shall support self-registration for Public Users to become (e.g., **Author, Registered Editor, …**), subject to administrative approval.
- **FR-UM-002:** System Administrators shall have the ability to create, modify, suspend, and delete user accounts and assign roles.
- **FR-UM-003:** User access to modules and features shall be strictly governed by their assigned role (RBAC), ensuring users can only perform actions they are authorized for.

- **FR-UM-004:** The system shall provide a secure "Forgot Password" workflow that allows users to reset their password via a link sent to their registered email address.
- **FR-UM-005:** All user sessions shall automatically time out after a configurable period of inactivity, requiring the user to log in again.

3.2.9 Citation Generation Tools

- **FR-CITE-001:** (On-Demand Citation Generation) The system shall provide a "Cite" or "Generate Citation" button on the main page for every journal article, published eBook, and item in the repository.
- **FR-CITE-002:** (Multiple Citation Styles) Upon clicking the "Cite" button, the system shall generate and display ready-to-copy citations in several major academic styles. At a minimum, the system shall support:
    - APA (American Psychological Association)
    - MLA (Modern Language Association)
    - Chicago (Author-Date)
    - Harvard
- **FR-CITE-003:** (Data Accuracy and Completeness) All generated citations must be accurate and automatically populated with the correct metadata from the system, including author(s), title, publication year, journal title, volume, issue, page numbers, and the permanent DOI link.
- **FR-CITE-004:** (User-Friendly Interface) The citation generation tool shall appear in a user-friendly modal window or pop-up. It must provide a simple "Copy to Clipboard" button for each citation style to allow for easy pasting into a user's document.
- **FR-CITE-005: (**Export for Reference Managers) The tool shall also provide options to export the citation data in standard formats compatible with popular reference management software. At a minimum, it must support export in RIS (.ris) and BibTeX (.bib) formats for use in EndNote, Zotero, Mendeley, and other similar programs.

3.3 Nonfunctional Requirements

Performance

- **Page Load Speed:** The average page load time for all publicly accessible pages shall be under 3 seconds on a standard broadband connection.
- **API Response Time:** The server's API response time for typical data queries shall be under 500 milliseconds.
- **Concurrent User Load:** The system must be engineered to support a minimum of 100 concurrent active users without significant performance degradation.

Scalability

- **Architectural Scalability:** The system's architecture will be horizontally scalable, allowing for the addition of more server instances to handle increased load as the user base grows.
- **Data Scalability:** The database design and queries will be optimized to efficiently manage a growing volume of data, projected at a 20% increase per year, without loss of performance.

Security

- **Password Protection:** All user passwords must be protected using a strong, one-way, salted hashing algorithm, such as bcrypt.
- **Vulnerability Mitigation:** The platform will be secured against common web vulnerabilities as defined by the most current OWASP Top 10 list, including SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).
- **Security Auditing:** A mandatory third-party penetration test and vulnerability assessment will be successfully completed before the platform's final public launch.

Reliability and Availability

- **System Uptime:** The platform will maintain a target uptime of 99.5%, calculated monthly, excluding pre-scheduled maintenance windows.
- **Data Backup**: Automated daily backups of the entire database and all user-uploaded files will be implemented and stored securely.
- **Disaster Recovery**: A formal disaster recovery plan will be in place to allow for the full restoration of service within 24 hours in the event of a catastrophic system failure.

Usability

- **Workflow Efficiency**: Key user workflows, such as user registration and article submission, shall be designed to be completed in the minimum number of steps and be intuitive for non-technical users.
- **System Feedback:** The system will provide clear and immediate feedback to users for all actions, including success messages, loading indicators, and understandable validation errors.

Maintainability

- **Code Quality:** The source code will be well-documented, follow consistent coding standards, and be organized logically to facilitate future development and bug fixing.
- **Modularity:** The system's modular architecture will allow for individual components (e.g., the Journal System) to be updated or maintained independently without impacting the rest of the platform.

Portability

- **Environment Independence:** The backend application will be containerized using Docker, ensuring it is portable and can be deployed consistently across different cloud providers or on-premise infrastructure that supports Docker.

3.4 Other Requirements

Comprehensive training shall be provided to ORA administrators and content managers to ensure a smooth handover and self-sufficient operation of the platform.

4 Analysis Models

This section provides a detailed analysis of the system's intended behavior through use cases. It defines the interactions between various actors and the ORA Digital Platform to achieve specific goals.

4.1 List of System Use Cases

This table provides a comprehensive list of all system use cases identified for the ORA Digital Platform.

Table 17 List of System Use Cases

S/N	Use Case ID	System Use Case Name	Use Case Description
1	UC-SYS-01	Register for a New User Account	A new user creates an account on the platform, which may require administrative approval.
2	UC-SYS-02	Log In to the System	An existing user authenticates to gain access to role-specific features.
3	UC-JMS-01	Submit a Manuscript for Publication	An Author submits their academic manuscript to begin the peer-review process.
4	UC-JMS-04	Submit a Peer Review	A Reviewer evaluates an assigned manuscript and submits confidential feedback.
5	UC-JMS-06	Make a Final Editorial Decision	The Editor-in-Chief makes the final publication decision on a manuscript.
6	UC-JMS-10	Publish an Article	A Journal Manager performs production steps to publish an accepted article.
7	UC-BPS-01	Submit a Book Manuscript	An Author submits a book-length manuscript for consideration.
8	UC-BPS-04	Manage Digital Production of an eBook	A Digital Content Manager converts an accepted manuscript into final formats.
9	UC-LMS-01	Search the Library Catalog	A user searches physical and digital library resources.
10	UC-LMS-02	Borrow a Physical Book	A Librarian records the checkout of a physical book for a Member.
11	UC-LMS-06	Catalog a New Physical Book	A Cataloger enters a physical book into the catalog system.
12	UC-OWP-01	Create a New Wikipedia Article	A Registered Editor creates a new article for the Oromo Wikipedia.

13	UC-OWP-03	Moderate and Manage Wiki Content	A Sysop reverts vandalism, deletes, or protects pages.
14	UC-RMS-01	Deposit a Scholarly Work	A researcher uploads their work (dataset, thesis) to the repository.
15	UC-RMS-02	Curate and Validate a Submission	A curator reviews and enriches metadata of a repository submission.
16	UC-RNP-01	Create and Manage a Professional Profile	A researcher sets up their public profile.
17	UC-RNP-03	Create and Manage a Research Group	A member creates a collaborative group; Moderator manages it.
18	UC-CITE-01	Generate a Formatted Citation	A user generates a ready-to-copy citation.
19	UC-CITE-02	Export Citation to Reference Manager	A user downloads citation (.ris,.bib) for Zotero/EndNote.

4.2 Use Case Diagram

The following diagram provides a high-level overview of the primary actors and their interactions with the major functional areas of the ORA Digital Platform. It illustrates the scope of the system and the relationships between different user roles. The generalization (inheritance) arrows indicate that higher-level roles inherit all the capabilities of the roles below them.

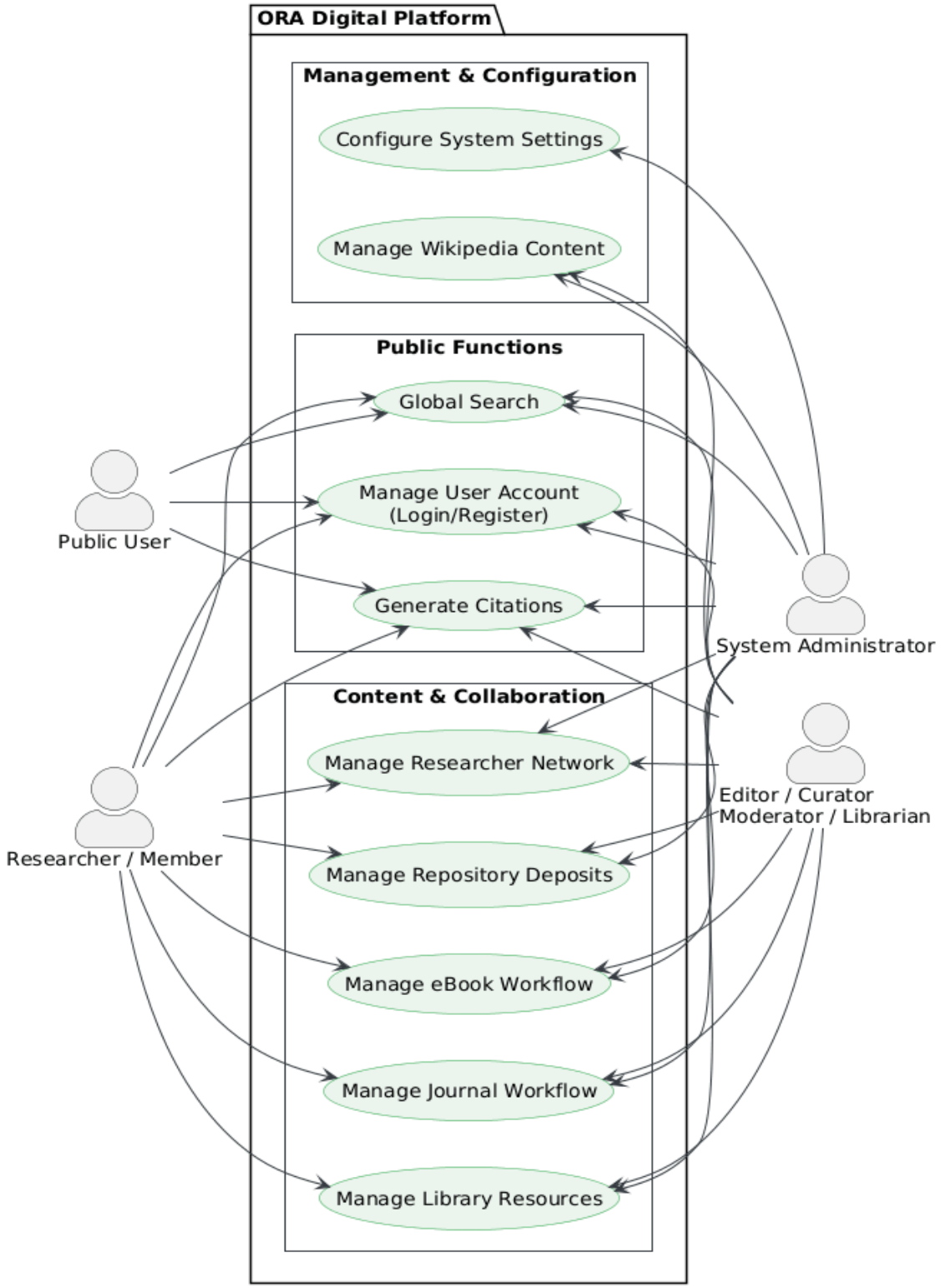

Figure 1 Use Case Diagram for ORA Digital Platform

4.3 System Use Case Detail

Table 18 Manage System User Registration Use case

Field	Description
Use Case ID	UC-SYS-01
Use Case Name	Manage System User Registration
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Public User
Description	This use case outlines the processes and procedures involved in the creation of a new user account on the ORA Digital Platform.
Trigger	When a user who is not logged in wants to create an account.
Preconditions	• The user has a valid, accessible email address. • The user's email address is not already registered in the system.
Normal Flow	1. System displays a page with a “Register” button or link. 2. Public User clicks “Register.” 3. System displays form requesting Name, Email, Password, Password Confirmation. 4. Public User enters all required fields. 5. User clicks “Create Account.” 6. System validates data (password match, email format...). 7. System saves user with **Pending Verification** status. 8. System sends verification link to user’s email.
Post Conditions	• A new System User record is created. • A verification email is sent to the user.
Alternative Flows	**3.1** User clicks **Cancel** → System discards input and returns to previous page.

Exceptions	**Exception 1: Data Integrity Issues** • *Description:* Required fields missing or inconsistent. • *Action:* Show error messages near invalid fields. **Exception 2: Duplicate Entries** • *Description:* Email already exists. • *Action:* Show message: "Account with this email already exists." **Exception 3: Data Validation Failures** • *Description:* Weak password, invalid email format, etc. • *Action:* Display specific validation errors. **Exception 4: System Errors** • *Description:* Server or email sending failure. • *Action:* Log error and show "Please try again later."
Priority	High
Effort/Complexity	Medium
Frequency of Use	High (especially during platform launch and growth)
Business Rules	• User account fields: Name, Email, Role, Status, Password (hashed). • Password must meet defined complexity rules. • Email must be unique. • Account must be email-verified before activation.
Special Requirements	• CAPTCHA mechanism to prevent bots.
Assumptions	• User has access to a valid email account. • User knows the platform URL.
Notes & TBD	• Define password complexity rules. • Set email verification link expiration time.
Related Misuse Case	• Malicious actor attempts bulk fake/spam registrations.

Sequence Diagram for: Manage System User Registration (UC-SYS-01)

This diagram shows the complete flow, including validation and email verification.

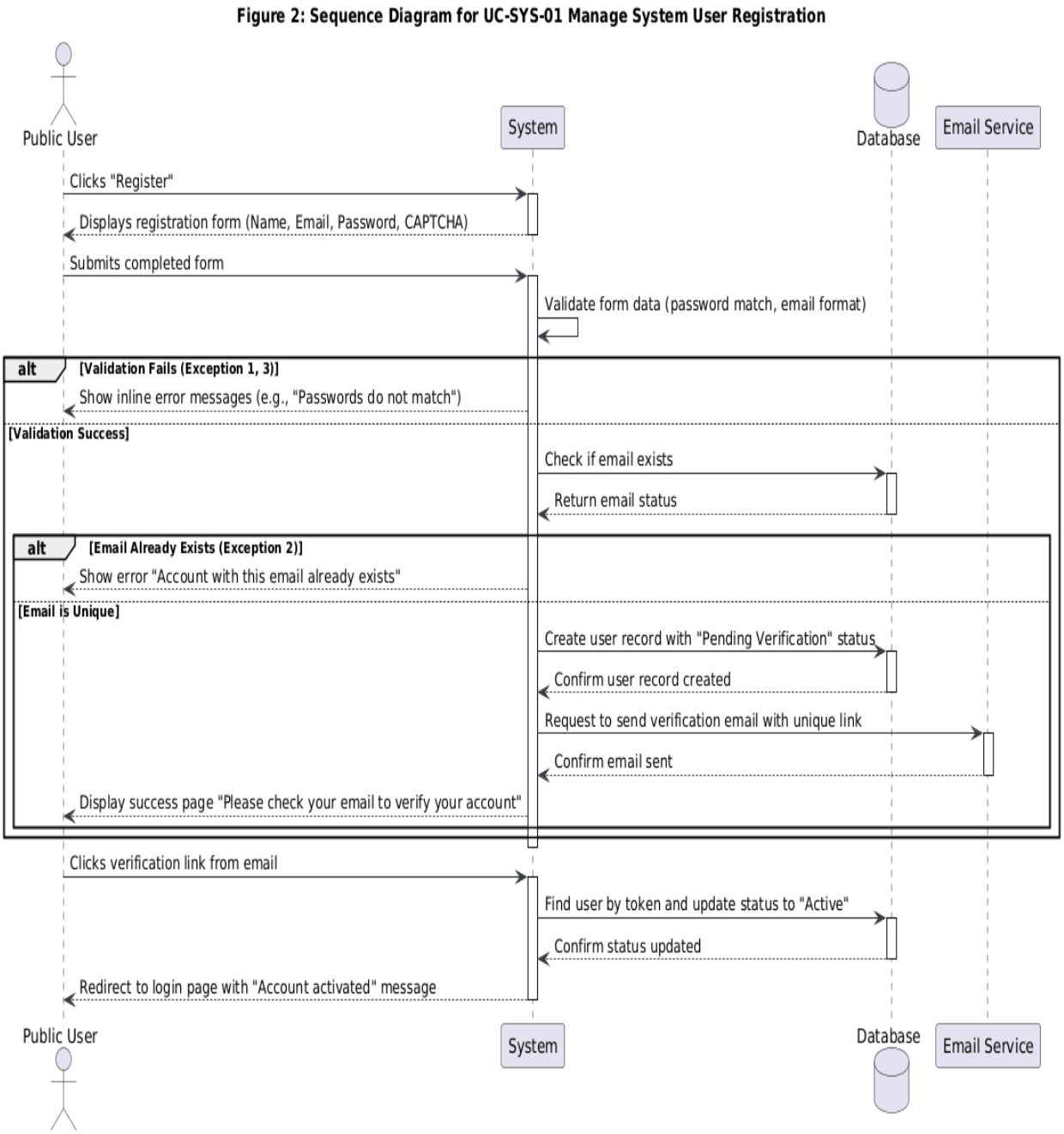

Figure 2 Sequence Diagram for: Manage System User Registration (UC-SYS-01)

Table 19 Manage User Login Use case

Field	Description
Use Case ID	UC-SYS-02
Use Case Name	Manage User Login
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Registered User (any user with an existing account)
Description	This use case outlines the process for an existing user to authenticate and gain access to the ORA Digital Platform.
Trigger	When a user wants to access features and content that require authentication.
Preconditions	• The user has a valid and active account (email verified). • Authentication service is operational.
Normal Flow	1. System displays a "Login" page. 2. User clicks "Login." 3. System displays login form (Email + Password). 4. User enters credentials. 5. User clicks "Login." 6. System validates credentials against database. 7. System checks that account is active and not locked. 8. System creates secure session and redirects user to dashboard/homepage.
Post Conditions	• Secure session is established. • User is granted access based on assigned roles and permissions.
Alternative Flows	**6.1 Invalid Credentials** 6.1.1 System shows: "Invalid email or password." 6.1.2 User stays on login page. **7.1 Account Not Active/Suspended** 7.1.1 System shows: "Your account is not active…" 7.1.2 User is denied access.
Exceptions	**Exception 1: Data Validation Failures** • Invalid email format, etc. • System shows inline validation errors. **Exception 2: System Errors** • Database or authentication failure. • System logs error and displays: "Please try again later."
Priority	High
Effort/Complexity	Low

Frequency of Use	Very High (every login attempt)
Business Rules	• Authentication must use HTTPS. • System must prevent brute-force attacks (e.g., lockouts). • After login, user role and permissions must be loaded into session.
Special Requirements	• Include “Forgot Password?” workflow. • Support “Remember Me” for extended sessions.
Assumptions	• User has valid login credentials. • User knows the platform URL.
Notes & TBD	• Define number of failed attempts before temporary lock. • Define “Remember Me” session duration.
Related Misuse Case	• Brute-force password guessing. • Login using stolen credentials.

Sequence Diagram for UC-SYS-02 Manage User Login

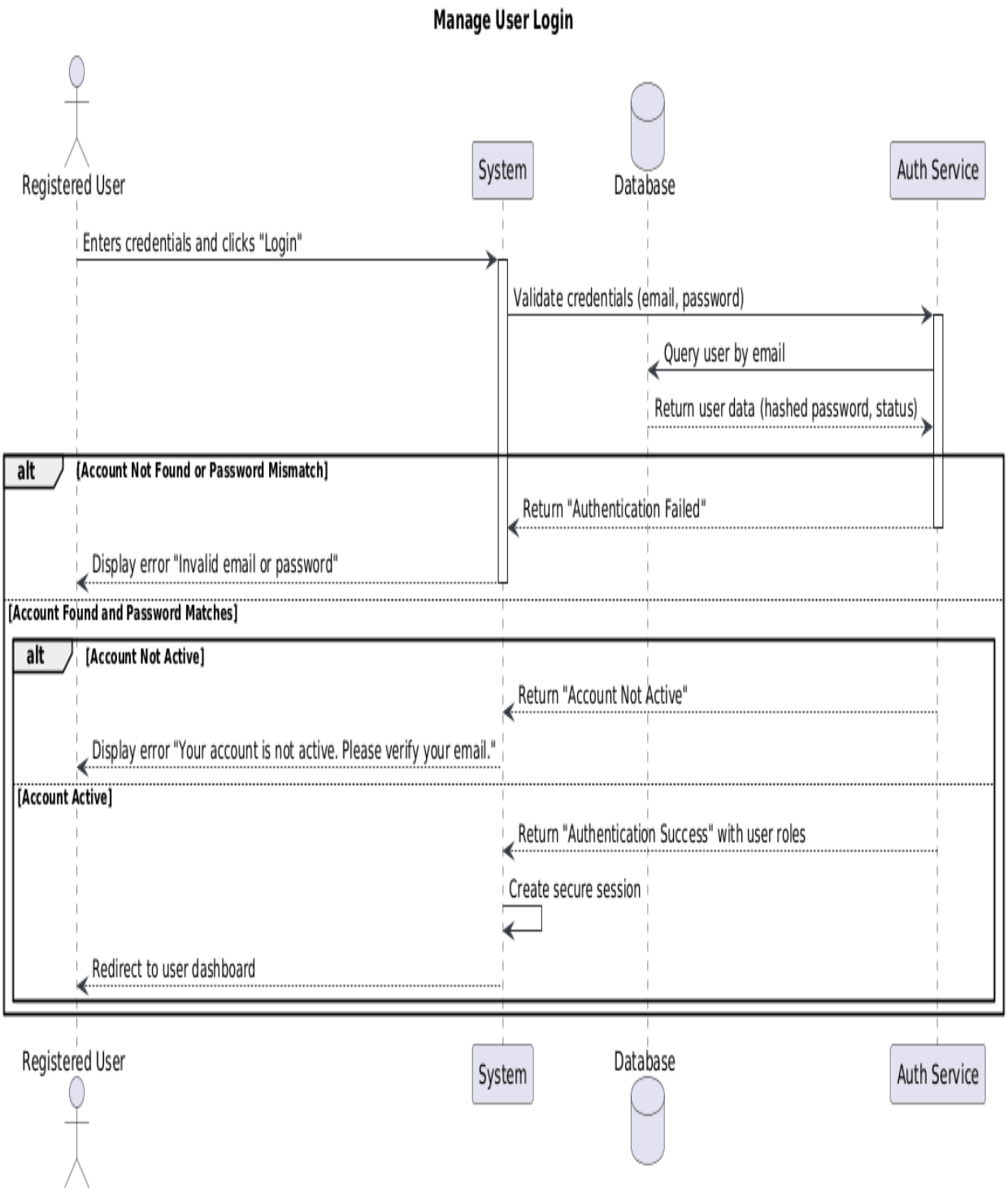

Figure 3 Sequence Diagram for UC-SYS-02 Manage User Login

Table 20 Manage Manuscript Submission Use case

Field	Description
Use Case ID	UC-JMS-01
Use Case Name	Submit a Manuscript for Publication
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Author
Description	Process for an Author to submit an academic manuscript for journal publication.
Trigger	The Author wishes to begin submitting a completed manuscript.
Preconditions	• User is logged in as **Author**. • Target journal is active and accepting submissions.
Normal Flow	1. System displays journal homepage. 2. Author selects "Submit Manuscript." 3. System shows multi-step submission form. 4. Author uploads manuscript and supplementary files. 5. Author enters metadata (title, abstract, keywords, co-authors). 6. Author enters cover letter & ethics statements. 7. System validates and saves the submission with a tracking ID.
Post Conditions	• Manuscript stored with **Submitted** status • Associate Editor notified.
Alternative Flows	2.1 Author clicks "Save as Draft." 2.1.1 System saves incomplete submission.
Exceptions	• Incorrect metadata. • Duplicate submission detected. • Unsupported file type or size limit exceeded.

	• System error during upload or save.
Priority	High
Effort/Complexity	Intermediate
Frequency of Use	High
Business Rules	• Title, abstract, and author required. • Only accepted file types allowed. • File size limits enforced. • All authors must agree to ethics statement.

Sequence Diagram for UC-JMS-01 Manage Manuscript Submission

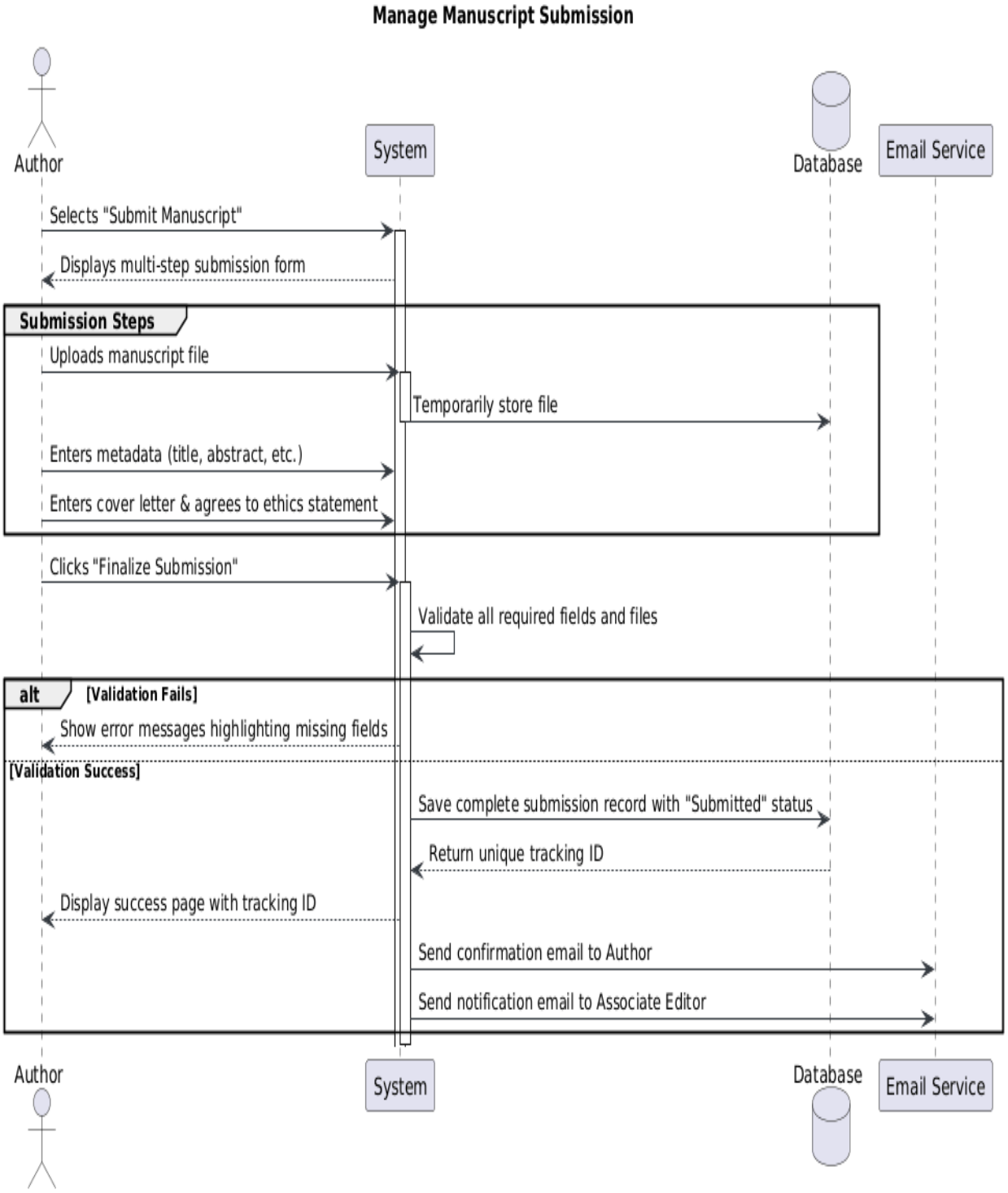

Figure 4 Sequence Diagram for UC-JMS-01 Manage Manuscript Submission

Table 21 Manage Peer Review Submission Use case

Field	Description
Use Case ID	UC-JMS-04
Use Case Name	Manage Peer Review Submission
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Reviewer
Description	This use case outlines the process for an assigned Reviewer to evaluate a manuscript and submit their confidential feedback and recommendation to the editorial team.
Trigger	When a Reviewer who has accepted an invitation is ready to submit their evaluation.
Preconditions	• User is authenticated with "Reviewer" role. • Reviewer has an assigned manuscript in "Under Review" state. • Review deadline has not passed.
Normal Flow	1. System shows Reviewer's dashboard with active assignments. 2. Reviewer selects a manuscript. 3. System shows review page with blinded manuscript download + review form. 4. Reviewer downloads and reads manuscript. 5. Reviewer fills the review form (editor-only comments & author comments). 6. Reviewer selects a final recommendation (Accept/Minor Revision/Major Revision/Reject). 7. Reviewer clicks **Submit Review**. 8. System validates required fields. 9. System saves review, links it to manuscript, and shows confirmation. 10. System notifies Associate Editor.
Post Conditions	• Reviewer's feedback is securely stored. • Manuscript review status updated (e.g., "1 of 2 Reviews Completed"). • Associate Editor is notified.
Alternative Flows	**5.1 Save Draft** 5.1.1 Reviewer clicks **Save Draft**. 5.1.2 System saves incomplete review for later completion.
Exceptions	**Exception 1: Data Integrity Issues** • Mandatory fields missing (e.g., recommendation). • System shows error and prevents submission. **Exception 2: Access Denied**

	• Reviewer attempts submission after deadline. • System blocks submission and advises contacting Editor. **Exception 3: System Errors** • Unexpected error saving data. • System logs error and shows friendly message.
Priority	High
Effort/Complexity	Medium
Frequency of Use	High (core part of peer-review workflow)
Business Rules	• System must maintain double-blind review: Reviewer ↔ Author identities hidden. • Reviewer comments for authors must not reveal identity. • Final recommendation is mandatory. • Review must be stored securely.
Special Requirements	• Automated email reminders must be sent as the deadline approaches.
Assumptions	• Reviewer has subject expertise. • Reviewer has stable internet for downloading and submitting reviews.
Notes & TBD	• Define full structure of the standard review form. • Define timing and content of reminder emails.
Related Misuse Case	• Reviewer submits intentionally biased/unprofessional feedback. • Reviewer tries to uncover author's identity.

Sequence Diagram for UC-JMS-04 Manage Peer Review Submission

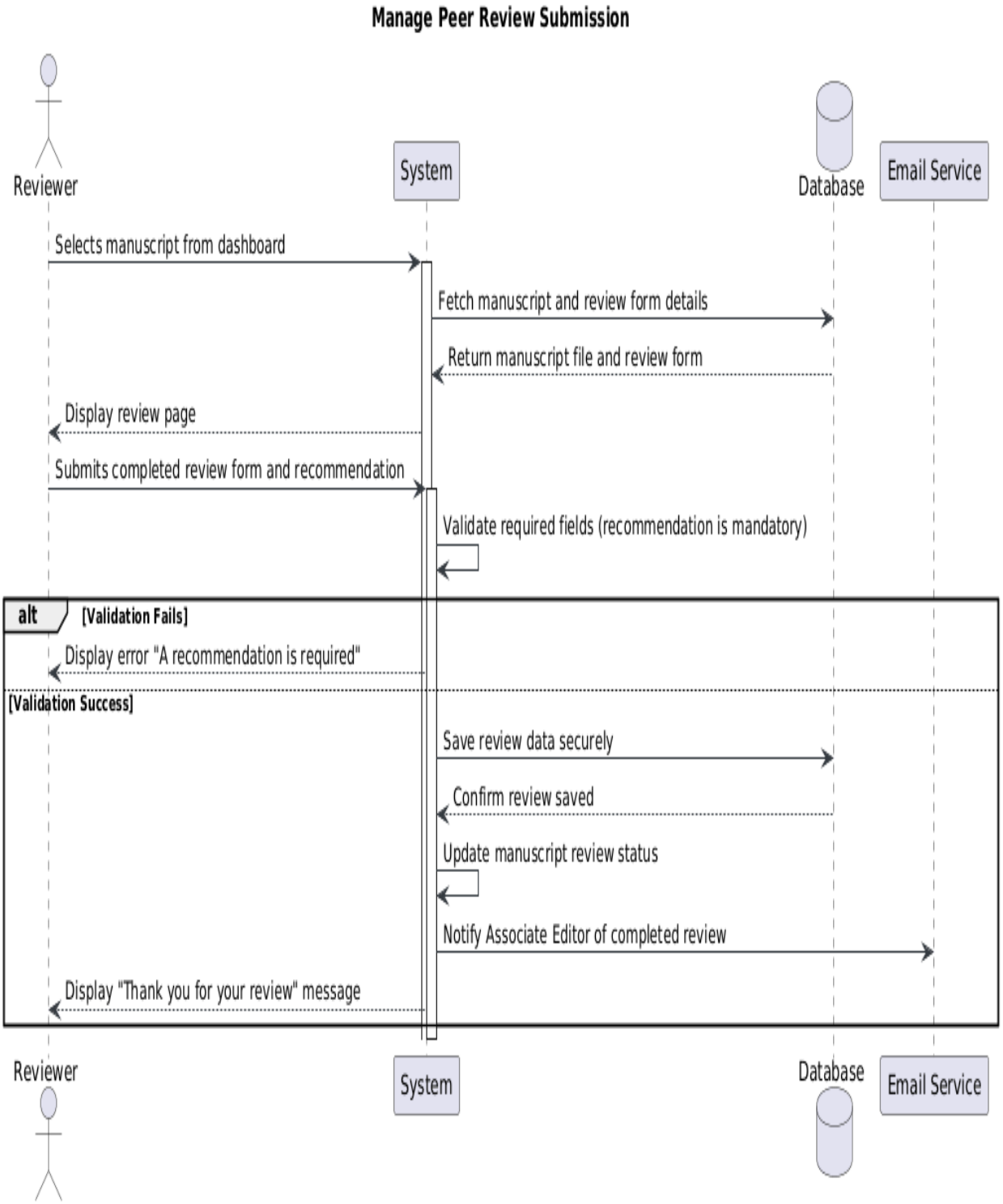

Figure 5 Sequence Diagram for UC-JMS-04 Manage Peer Review Submission

Table 22 Manage Editorial Decision Use case

Field	Description
Use Case ID	UC-JMS-06
Use Case Name	Manage Editorial Decision
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Editor-in-Chief (EIC), Associate Editor
Description	This use case describes the process for the Editor-in-Chief (EIC) to make the final publication decision on a manuscript after completing peer reviews and receiving the Associate Editor's recommendation.
Trigger	The EIC receives a notification that a manuscript recommendation is ready for final decision.
Preconditions	• EIC is authenticated and logged in. • Manuscript status is "Recommendation Submitted" or "Reviews Completed." • Required number of peer reviews have been submitted. • Associate Editor has submitted their recommendation.
Normal Flow	1. System displays the EIC dashboard showing manuscripts awaiting final decision. 2. EIC selects the manuscript. 3. System displays the decision page with: a. Manuscript file. b. Reviewer comments (confidential & author-visible). c. Associate Editor's summary & recommendation. 4. EIC clicks **"Make a Decision."** 5. System displays decision options (Accept, Minor Revision, Major Revision, Reject). 6. EIC selects a decision. 7. System opens a text editor with a decision letter template. 8. EIC finalizes the letter and clicks **"Submit Final Decision."** 9. System validates, updates manuscript status (e.g., Accepted), and logs the decision. 10. System emails the decision letter to the Author and notifies the Associate Editor.
Postconditions	• Manuscript status is updated based on the final decision. • Decision and letter are archived permanently. • Author is notified.
Alternative Flows	**6.1 Revisions Required**: 6.1.1. EIC selects Minor/Major Revision.

	6.1.2. System updates status to **"Revisions Required"** and notifies the Author to submit a revised version. **4.1 Clarification Required**: 4.1.1. EIC sends manuscript back to Associate Editor with a private note requesting additional information or a revised recommendation.
Exceptions	**Exception 1 – Premature Decision Attempt:** • If reviews or AE recommendation are missing, system disables decision options and displays pending items. **Exception 2 – System Error:** • If email sending or database save fails, system logs error, shows user-friendly error message, and preserves drafted decision letter.
Priority	High
Effort/Complexity	Medium
Frequency of Use	High (central to editorial workflow)
Business Rules	• Only the EIC can make final decisions. • Final decision allowed only after full review process. • All decision materials must be archived for audit. • Decision letters must not reveal reviewer identities.
Special Requirements	• Customizable email templates for each decision type. • Clear separation of reviewer comments vs. AE recommendation in the interface.
Assumptions	• EIC has authority on all editorial decisions. • EIC has fully reviewed all materials before deciding.
Notes / TBD	• Finalize decision letter templates. • Decide whether EIC can invite additional "tie-breaker" reviewers.
Related Misuse Case	• Unauthorized user attempts accessing decision interface. • EIC skips reviewing feedback and makes uninformed decision (process violation).

Sequence Diagram for UC-JMS-06 Manage Editorial Decision

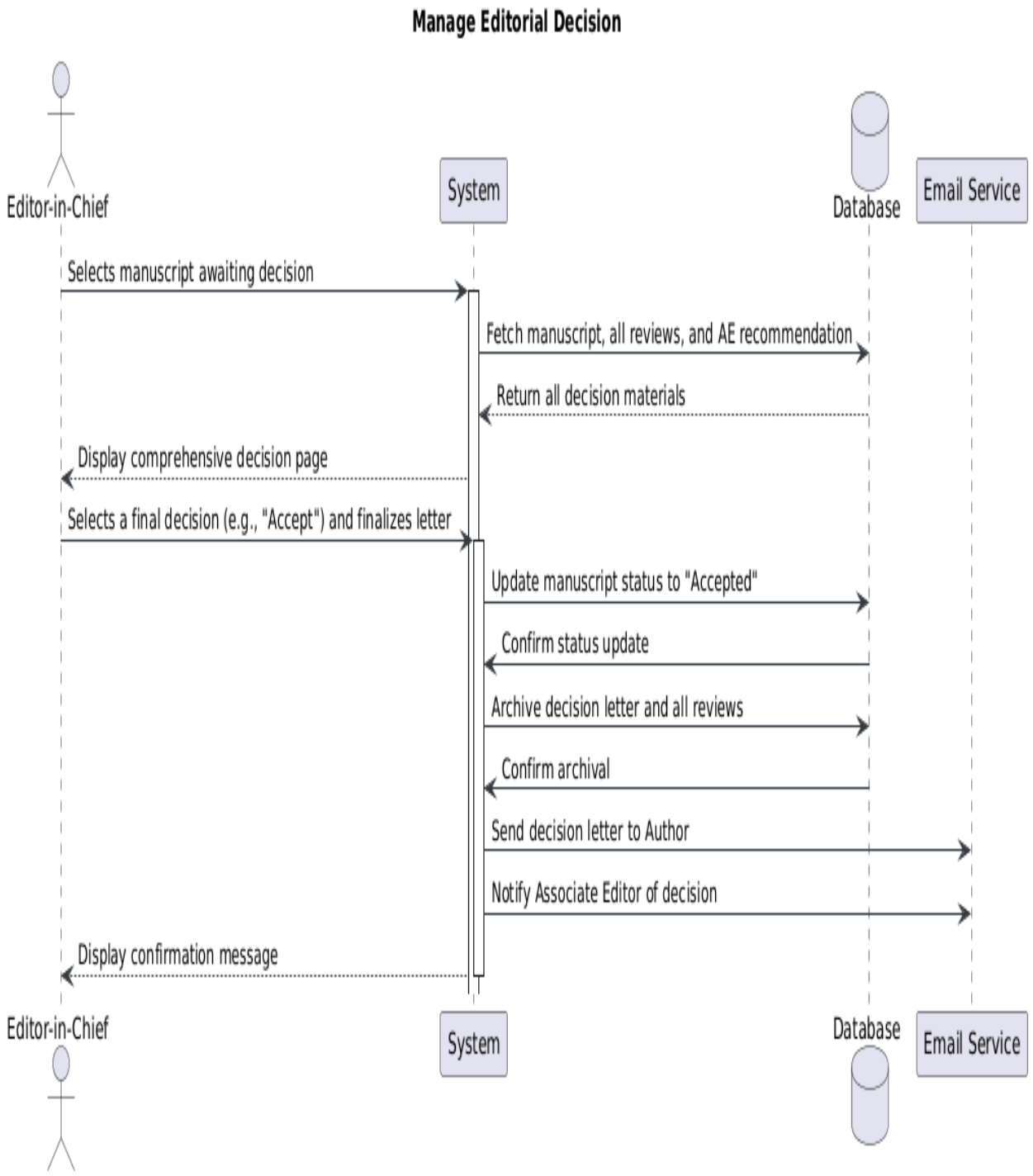

Figure 6 Sequence Diagram for UC-JMS-06 Manage Editorial Decision

Table 23 Manage Article Publication Use case

Field	Description
Use Case ID	UC-JMS-10
Use Case Name	Manage Article Publication
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Journal Manager, Author, System Administrator
Description	This use case describes the end-to-end production and publication process for an article that has received final acceptance from the Editor-in-Chief.
Trigger	Manuscript status is updated to **"Accepted."**
Preconditions	• User is authenticated with **Journal Manager** or **System Administrator** role. • Manuscript exists in **Accepted** status.
Normal Flow	1. System automatically notifies the Journal Manager that a manuscript is ready for production. 2. Journal Manager opens the manuscript in their production queue. 3. Journal Manager performs copyediting, language polishing, and reference checking. 4. Journal Manager typesets the edited manuscript and generates **PDF** and **HTML galley proofs**. 5. Journal Manager uploads the galley proof and sends it to the Author for proofreading. 6. Author reviews and either approves or requests corrections. 7. Journal Manager applies final corrections and prepares the publication-ready files. 8. Journal Manager opens the **Publish Article** screen. 9. Journal Manager assigns the article to a **specific journal, volume, and issue**. 10. Journal Manager initiates DOI generation. 11. System communicates with DOI agency and assigns a permanent DOI. 12. Journal Manager clicks **"Publish Online."** 13. System validates, publishes the article, and updates its status to **Published**.
Postconditions	• Article is published and publicly accessible. • Article is associated with a journal, volume, and issue. • DOI is permanently registered and attached to the article.

Alternative Flows	**6.1 Author Requests Corrections**: 6.1.1. Author submits minor corrections. 6.1.2. Journal Manager applies changes and may resend updated proof for confirmation. **11.1 Scheduled Future Publication**: 11.1.1. Journal Manager sets a future/embargo publication date. 11.1.2. System auto-publishes on that date and keeps article private until then.
Exceptions	**Exception 1 – DOI Generation Failure**: • DOI agency unreachable. • System logs error, shows message, saves publication as draft, and allows retry. **Exception 2 – Author Unresponsive**: • Author does not approve proof within allowed time. • System sends reminders and escalates to Journal Manager and EIC. **Exception 3 – Publication System Error**: • System fails during publishing. • System prevents partial publishing, logs error, and notifies System Administrator.
Priority	High
Effort/Complexity	Medium
Frequency of Use	High (applies to every accepted article)
Business Rules	• Article cannot be published without Author's final proof approval. • DOI must be assigned before public release. • Once published, content and DOI cannot be altered; corrections require formal **errata** workflow. • All published articles must belong to a **journal → volume → issue** hierarchy.
Special Requirements	• Integration with DOI registration agency (e.g., Crossref). • System must generate production-quality **PDF** and **HTML** output using predefined templates.
Assumptions	• Journal Manager is skilled in editing and typesetting. • ORA has an active DOI registration account.
Notes and TBD	• Define Author proofreading deadline and reminder schedule. • Finalize design templates for PDF and HTML formats.
Related Misuse Case	• Unauthorized user attempts to access publication actions. • Attempt to publish a manuscript that is not in "Accepted" state.

Sequence Diagram for UC-JMS-10 Manage Article Publication

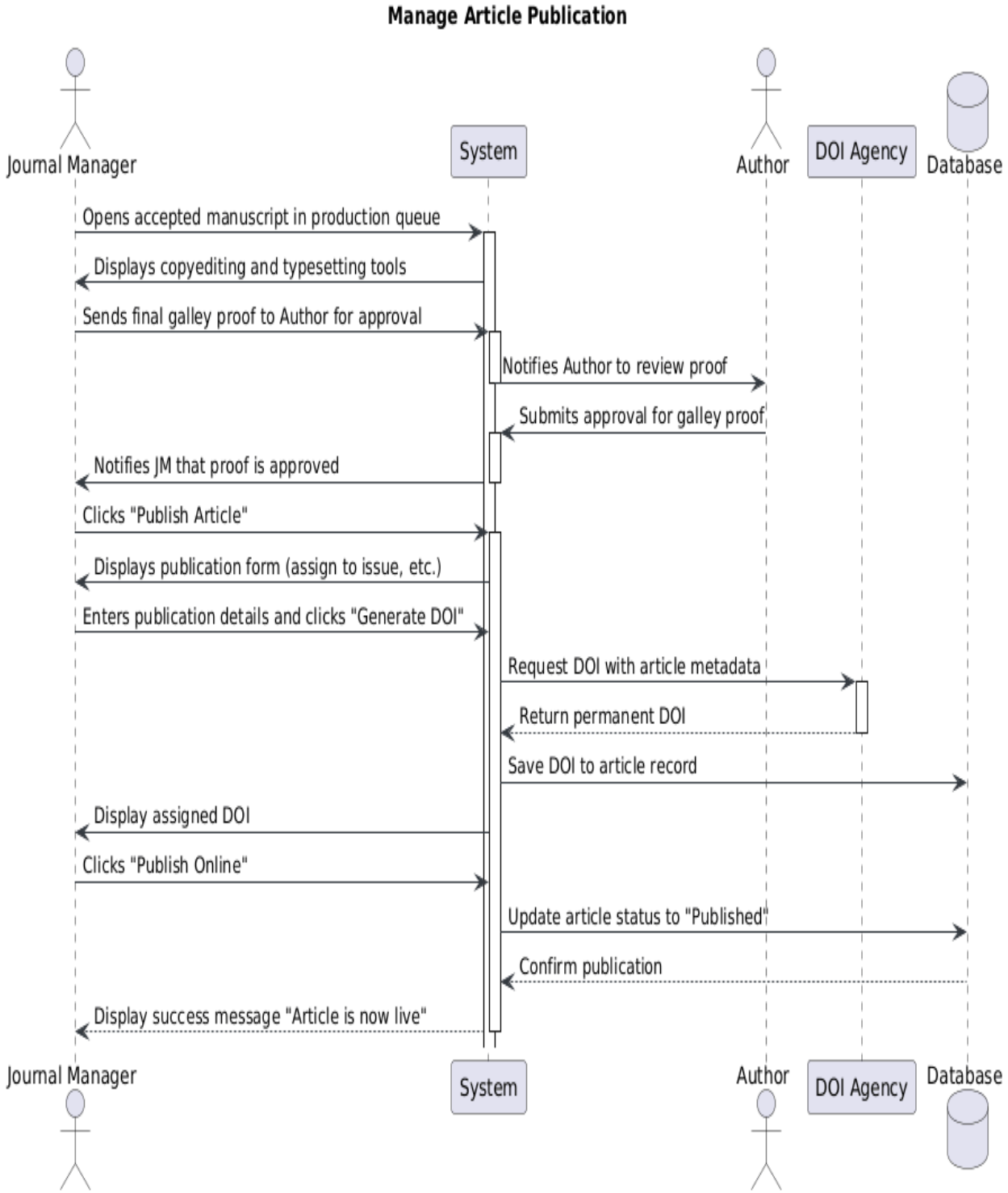

Figure 7 Sequence Diagram for UC-JMS-10 Manage Article Publication

Table 24 Manage Book Submission Use case

Field	Description
Use Case ID	UC-BPS-01
Use Case Name	Manage Book Submission
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Author, Researcher
Description	This use case describes the process for submitting a full book manuscript—including metadata and contributor information—to the ORA eBook Publishing System for initial screening and consideration.
Trigger	The Author is ready to submit a complete book-length manuscript.
Preconditions	• User is authenticated with **Author** or **Researcher** role. • The Author has the complete manuscript file (e.g., PDF, DOCX) ready for upload.
Normal Flow	1. System displays the Author's dashboard. 2. Author navigates to **eBook Publishing → Submit New Book**. 3. System displays the book submission form. 4. Author uploads the primary manuscript file. 5. Author enters required metadata: title, abstract/synopsis, keywords, proposed table of contents. 6. Author provides contributor information (authors/editors, affiliations). 7. Author reviews the submission and clicks **Submit for Consideration**. 8. System validates the form and checks the manuscript file. 9. System saves the submission, assigns a **unique tracking ID**, and sets its status to **Submitted**. 10. System displays a confirmation message and sends a confirmation email to the Author. 11. System notifies the assigned Book Editor about the new submission.
Postconditions	• A new book submission with **Submitted** status is created. • Book Editor is notified and can start the initial screening process.
Alternative Flows	**5.1 Save as Draft**: 5.1.1. Author clicks **Save Draft** at any point. 5.1.2. System saves the draft so the Author can complete it later.
Exceptions	**Exception 1 – Missing Mandatory Fields**: • Mandatory metadata (e.g., Title) missing. • System prevents submission and shows validation errors. **Exception 2 – File Upload Error**: • File exceeds size limit or unsupported format.

	• System shows a specific error and allows retry. **Exception 3 – System Error**: • Database or server error during save. • System logs the error and attempts to preserve entered data.
Priority	High
Effort/Complexity	Intermediate
Frequency of Use	Medium (less frequent than article submissions, but essential to eBook publishing)
Business Rules	• Every book submission must include a primary manuscript file. • Title and at least one author are mandatory. • File size limits must be enforced. • Once submitted, the Author cannot edit the book submission while under review.
Special Requirements	• System must reliably support large file uploads (e.g., long manuscripts).
Assumptions	• Author follows ORA formatting guidelines. • Author has the legal right to submit the manuscript.
Notes / TBD	• Finalize list of mandatory metadata fields. • Determine maximum file size for uploads.
Related Misuse Case	• Submission of plagiarized content. • Submission of spam or irrelevant files disguised as manuscripts.

Sequence Diagram for UC-BPS-01 Manage Book Submission

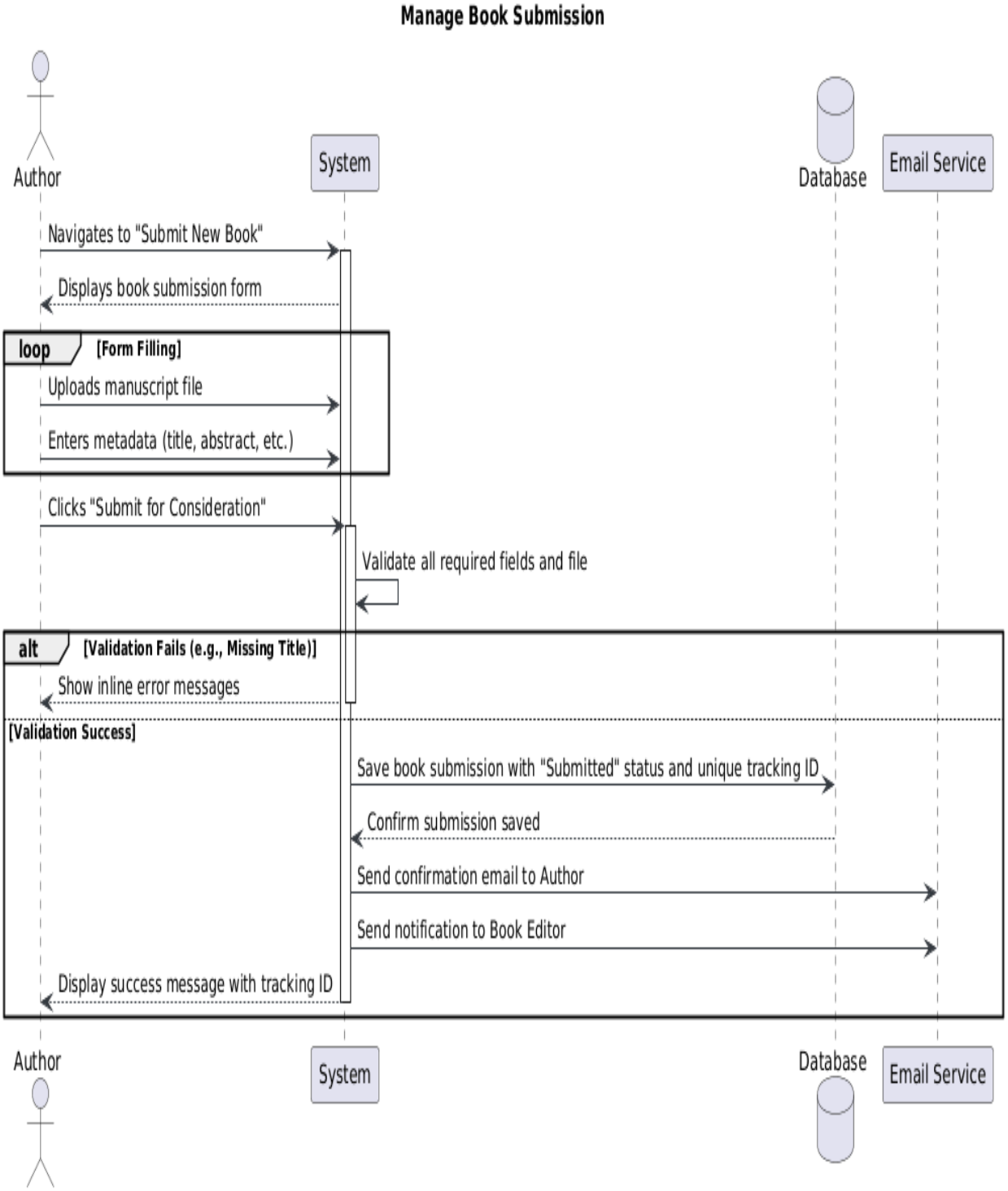

Figure 8 Sequence Diagram for UC-BPS-01 Manage Book Submission

Table 25 Manage eBook Production Use case

Field	Description
Use Case ID	UC-BPS-09
Use Case Name	Manage eBook Production
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Digital Content Manager
Description	This use case describes the technical and administrative steps required to convert an accepted and financially cleared book manuscript into a final, publishable eBook (PDF/EPUB) with all necessary metadata.
Trigger	The Digital Content Manager receives a system notification that a manuscript has been editorially accepted and financially cleared.
Preconditions	• User is authenticated with the **Digital Content Manager** role. • Book manuscript exists with **Accepted** status. • Manuscript has **Financial Clearance: Approved** status.
Normal Flow	1. System displays the Digital Content Manager's dashboard, highlighting manuscripts in the **Production Queue**. 2. Digital Content Manager selects a manuscript to begin production. 3. Downloads the final manuscript file. 4. Performs final proofreading and layout quality check. 5. Converts the manuscript into eBook formats (PDF, EPUB). 6. Assigns an official ISBN from ORA's registered block. 7. Initiates DOI generation; system communicates with the DOI agency to assign a permanent DOI. 8. Uploads final eBook files to the ORA digital repository. 9. Enters/verifies final metadata (ISBN, DOI, author, keywords, etc.) in the Library Management System. 10. Sets final access rights (Open Access, Restricted, Embargoed). 11. Updates the book's status to **Ready for Publication** or **Published**.
Postconditions	• Publication-ready eBook files (PDF, EPUB) are stored in the repository. • Official ISBN and DOI are registered and linked to the book. • Book status is updated and ready for public release.
Alternative Flows	**4.1 Layout Errors Found**: 4.1.1. Digital Content Manager detects significant formatting or layout issues. 4.1.2. Changes manuscript status to **Production Halted** and returns it to the Book Editor with correction notes.

Exceptions	**Exception 1 – Missing Identifiers**: • ISBN or DOI missing. • System prevents completion and displays: "Cannot complete production. ISBN and DOI must be assigned." **Exception 2 – File Conversion Failure**: • System fails to generate PDF/EPUB. • System logs the error and notifies the Digital Content Manager to check the source file. **Exception 3 – DOI Generation Failure**: • System cannot connect to DOI agency API. • Logs the error, shows a message, and allows progress to be saved for later retry.
Priority	High
Effort/Complexity	High
Frequency of Use	Medium (occurs for every accepted book manuscript)
Business Rules	• Production cannot start without financial clearance. • ISBN is mandatory for every published eBook. • Every eBook must have a unique DOI. • eBook files must pass internal QA before publication.
Special Requirements	• System must convert documents into professional-quality PDF and EPUB. • Integrated with DOI registration API. • Secure module for ISBN assignment.
Assumptions	• Digital Content Manager has required skills/software. • ORA has an official ISBN block. • ORA has an active DOI account.
Notes / TBD	• Define QA checklist for final eBook files. • Determine ISSN assignment if part of a series.
Related Misuse Case	• Unauthorized access to production module. • Duplicate ISBN assignment.

Sequence Diagram for UC-BPS-09 Manage eBook Production

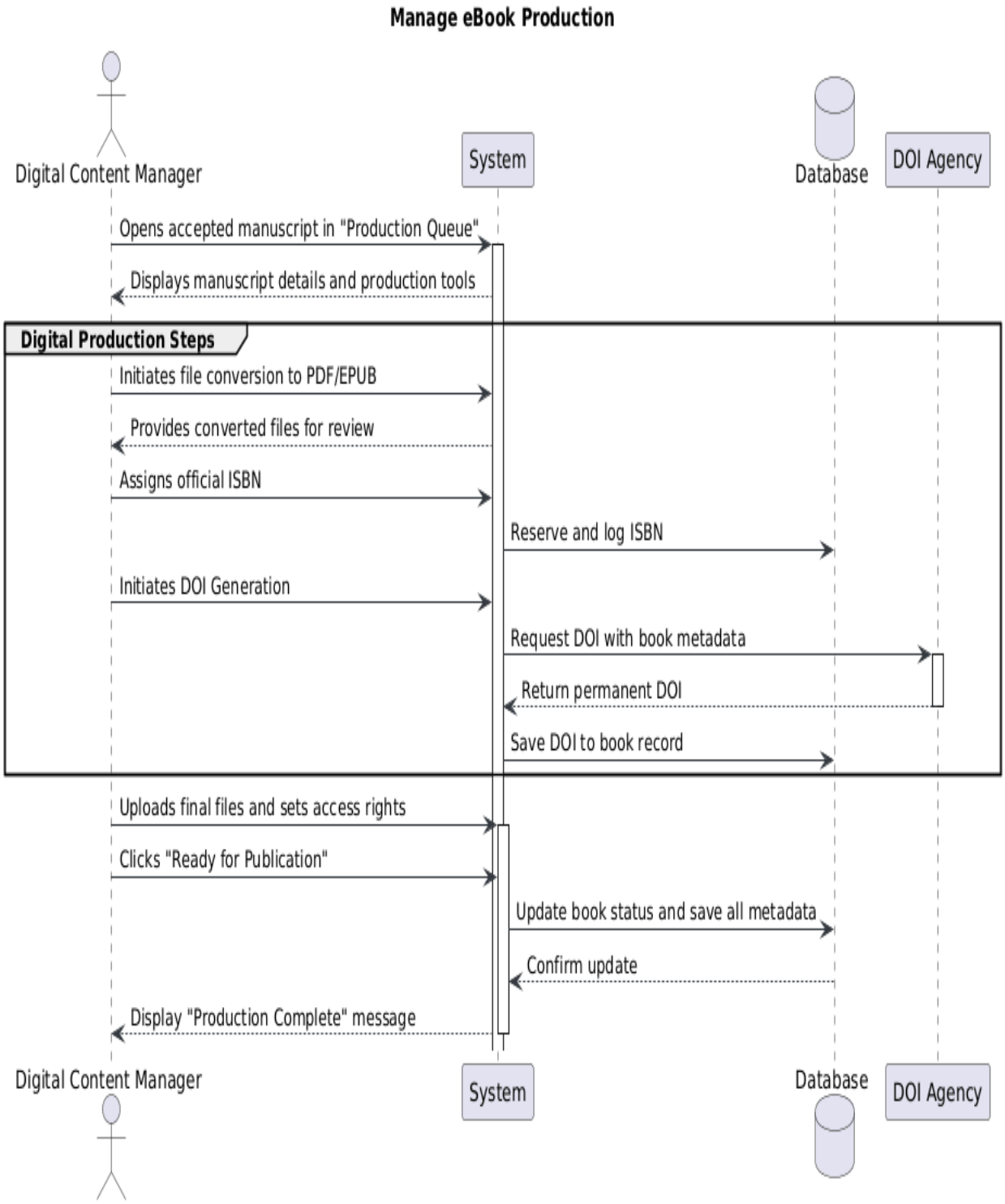

Figure 9 Sequence Diagram for UC-BPS-09 Manage eBook Production

Table 26 Search Library Catalog Use case

Field	Description
Use Case ID	UC-LMS-01
Use Case Name	Search Library Catalog
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Any User (Public User, Member, Student, etc.)
Description	This use case describes how a user searches the ORA Library's Online Public Access Catalog (OPAC) to discover and locate physical or digital library resources.
Trigger	When a user wants to find a specific item or browse for resources on a particular topic.
Preconditions	• The Library Management System (LMS) is online. • The catalog contains at least one indexed item.
Normal Flow	1. User navigates to the "Library" section of the ORA platform. 2. System displays the main library page with a prominent search bar. 3. User enters a search query (title, author, keyword, etc.). 4. User initiates the search by clicking **Search** or pressing **Enter**. 5. System processes the query against the catalog index. 6. System displays a results page listing all matching items. 7. Each result displays: Title, Author, Format (Physical Book, eBook, Journal, etc.), Availability Status (Available, Checked Out). 8. User selects an item by clicking its Title. 9. System displays the full item record with complete bibliographic information and related actions.
Postconditions	• User is presented with items relevant to their query. • User may proceed to "Access Digital Resource," "Place a Hold," "Reserve Item," or related use cases.
Alternative Flows	**6.1 No Results Found** 6.1.1. System displays message: "No results found for your search." 6.1.2. System suggests spelling corrections or alternative keywords. **3.1 Advanced Search** 3.1.1. User clicks **Advanced Search**. 3.1.2. System displays fields such as Title, Author, Subject, ISBN, Publication Year, Format. 3.1.3. User fills one or more fields and performs search. 3.1.4. System continues from Normal Flow step 6.

	6.2 Filter and Sort 6.2.1. User applies filters (e.g., Format: eBook; Language: Afan Oromo; Year: 2020+). 6.2.2. System refreshes result instantly. 6.2.3. User may choose sort options such as Relevance or Publication Date.
Exceptions	**Exception 1: System Error•** Description: System experiences unexpected issue querying the index/database. • Action: System logs the error and shows: "Search service temporarily unavailable. Please try again later."
Priority	Very High
Effort / Complexity	Medium
Frequency of Use	Very High (primary user interaction in LMS)
Business Rules	• Basic search must query Title, Author, Subject/Keywords fields. • Search must be case-insensitive. • Real-time availability must be shown for physical items. • Search results must use pagination (e.g., 20 items per page).
Special Requirements	• Search must return results within acceptable performance thresholds. • System should use search indexing for accuracy and relevance ranking. • Filters should update results dynamically.
Assumptions	• User has basic knowledge of the item being searched (keywords, title, etc.).
Notes / TBD	• Finalize list of filter categories (Format, Language, Genre, Year, etc.). • Determine default sort order (Relevance vs. Publication Date).
Related Misuse Cases	• Malicious user performs a high-frequency query attack (DoS attempt). • User attempts SQL injection via search input.

Sequence Diagram for UC-LMS-01 Search Library Catalog

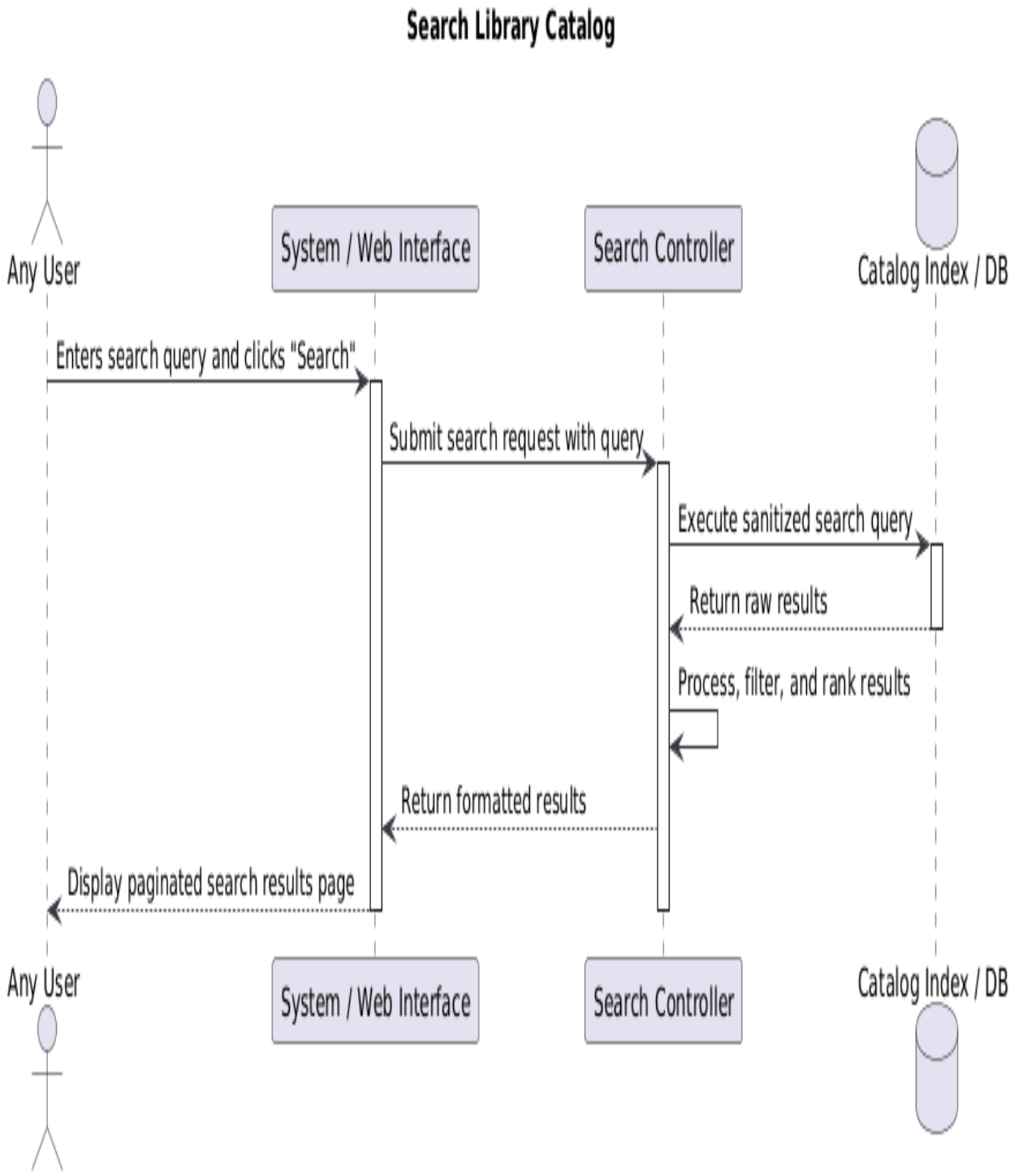

Figure 10 Sequence Diagram for UC-LMS-01 Search Library Catalog

Table 27 Manage Book Lending Use case

Field	Description
Use Case ID	UC-LMS-02
Use Case Name	Manage Book Lending
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Librarian (Physical), Member / User / Student
Description	This use case describes how a Librarian checks out physical books to Members and checks them back in, ensuring accurate loan records, fine calculation, and real-time inventory updates.
Trigger	A Member brings one or more physical books to the circulation desk for borrowing or returning.
Preconditions	• Librarian is logged in with **Librarian (Physical)** role. • Member has an active library account. • The book has a valid barcode and exists in the catalog.
Normal Flow (Check-Out)	1. Librarian opens the **Circulation** module. 2. System loads the circulation interface. 3. Librarian scans the Member's library card or searches for the Member manually. 4. System displays the Member's account details and borrowing eligibility. 5. Librarian scans the barcode of the book to be borrowed. 6. System validates borrowing rules (borrow limit, no blocks, book availability). 7. System creates a new loan record tying the book to the Member. 8. System calculates and displays the due date based on circulation policies. 9. Librarian confirms the transaction. 10. System updates the book's status to **"Checked Out."** 11. Librarian informs the Member of the due date.
Postconditions	• **Check-Out:** Loan record created; book status = *Checked Out*. • **Check-In:** Loan record closed; book status = *Available* or *On Hold*.
Alternative Flows	**6.1 Member Account Blocked** 6.1.1. System displays an alert (e.g., unpaid fines, suspended account). 6.1.2. Librarian informs the Member; loan is denied. **6.2 Book Not Eligible for Loan** 6.2.1. System shows error (e.g., Reserved, On Hold, Reference-Only). 6.2.2. Loan is denied.

	1.1 Check-In Process 1.1.1. Librarian scans a returned book. 1.1.2. System retrieves and closes the active loan record. 1.1.3. System checks overdue status. 1.1.4. If overdue, system calculates fines and adds them to Member's account. 1.1.5. System updates status to **Available** OR **On Hold** (if reserved).
Exceptions	**Exception 1: Invalid Barcode** • Barcode not found in system. • System displays: **"Item not found."** Librarian must search manually or re-catalog item. **Exception 2: System Error** • Connection or database error occurs during transaction. • System cancels the transaction, logs the issue, and notifies the Librarian to retry.
Priority	Very High
Effort / Complexity	Medium
Frequency of Use	Very High (core daily circulation task)
Business Rules	• Book status must update in real-time. • Borrow limits must be strictly enforced. • Overdue fines calculated automatically based on configurable rules. • Returned books with active holds must move to **Hold Shelf**, not general availability. • Loan periods may vary by user type (students, staff, external users).
Special Requirements	• Must support standard USB barcode scanners. • UI must be optimized for fast transactions to handle long queues. • System must log all circulation activities for audit.
Assumptions	• All items and Members are already registered in the library system. • The library circulation desk has required hardware (computer, scanner).
Notes / TBD	• Define loan period for each user group. • Configure fine rates and maximum overdue limits. • Add workflow for lost/damaged items (replacement fee, waiver process).
Related Misuse Cases	• Borrow attempt using another user's card. • Librarian overrides account blocks without authorization.

Sequence Diagram for UC-LMS-02 Manage Book Lending

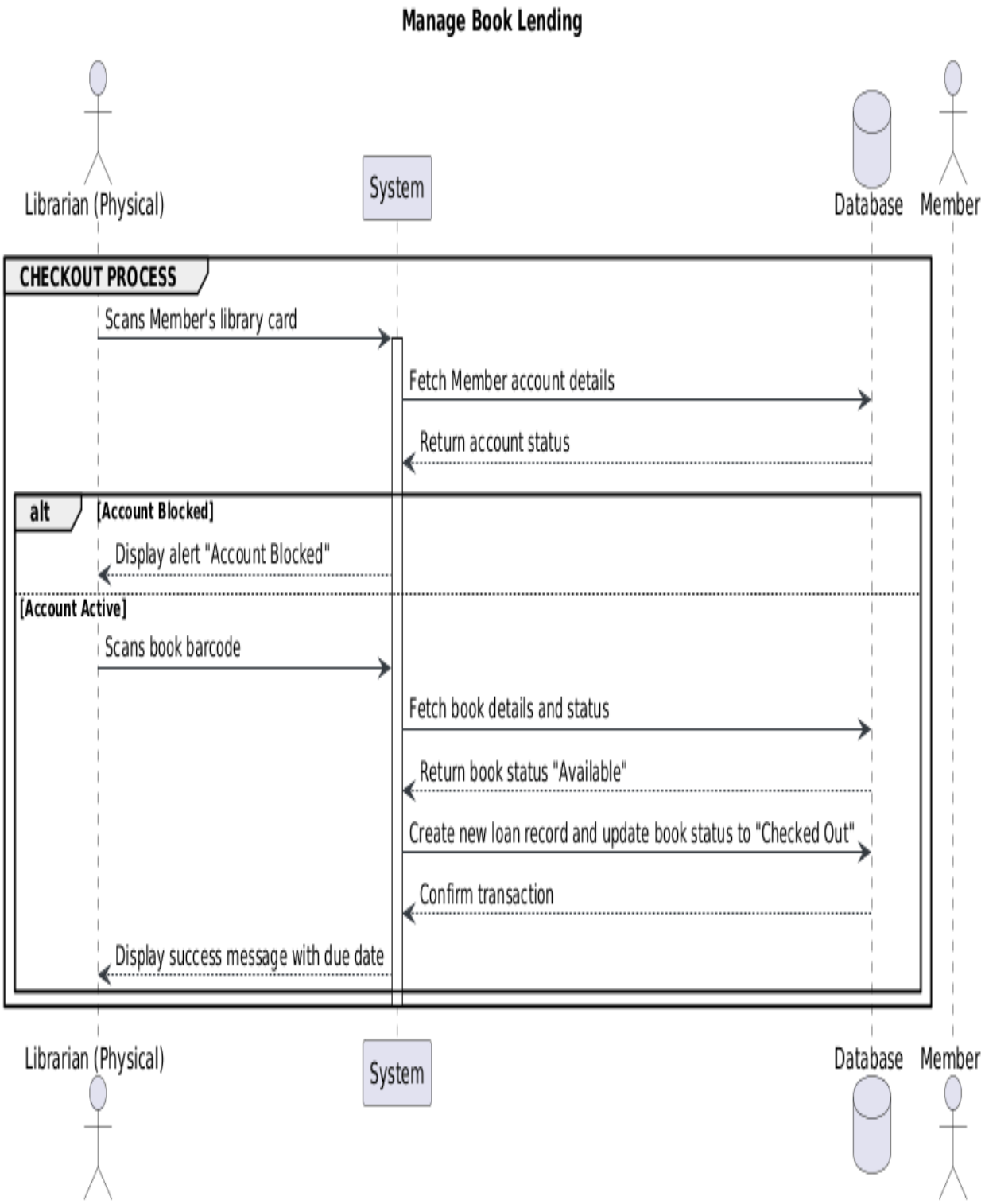

Figure 11 Sequence Diagram for UC-LMS-02 Manage Book Lending

Table 28 Manage Library Cataloging Use case

Field	Description
Use Case ID	UC-LMS-12
Use Case Name	Manage Library Cataloging
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Cataloger, Digital Librarian
Description	Describes the process of creating, updating, and maintaining bibliographic records for physical and digital resources to ensure they are accurate, complete, and discoverable in the Library Management System.
Trigger	A new resource is acquired or an existing record requires modification.
Preconditions	• User is authenticated with the **Cataloger** or **Digital Librarian** role. • The resource or its metadata is available for entry.
Normal Flow (Creating a New Record – Physical Book)	1. Cataloger logs in and opens the **Cataloging Module**. 2. Cataloger selects **"Create New Item Record."** 3. System displays a standard cataloging template. 4. Cataloger enters bibliographic details (Title, Author, Publisher, ISBN, etc.). 5. Cataloger applies classification (DDC/LCC) and assigns a Call Number. 6. Cataloger assigns a **unique barcode** to the item. 7. Cataloger adds subject headings and keywords. 8. Cataloger clicks **Save Record**. 9. System validates mandatory fields and checks for duplicates. 10. System saves the record and makes it discoverable in OPAC.
Postconditions	• A complete, validated bibliographic record is stored and searchable. • Updated records reflect accurate information.
Alternative Flows	**AF-1 Modify Existing Record** 1. Cataloger searches the record. 2. Opens it and selects **Edit**. 3. Makes changes and saves. 4. System updates the record. **AF-2 Import via ISBN** 1. Cataloger enters ISBN in the form.

	2. Clicks **Import Metadata**. 3. System fetches metadata from an external bibliographic source. 4. Cataloger reviews, adjusts, and continues from Step 5 of the normal flow.
Exceptions	**Exception 1: Missing Mandatory Fields** • System blocks saving and highlights required fields. **Exception 2: Duplicate Record Detected** • System warns user and gives options to edit existing or proceed for different edition. **Exception 3: System/Database Errors** • System logs error and shows "Unable to save. Try again later."
Priority	High
Effort/Complexity	High
Frequency of Use	High (daily cataloging operations)
Business Rules	• Each catalog entry must have a **unique identifier**. • Physical items must have **unique barcodes**. • Cataloging form must follow bibliographic standards (e.g., MARC). • All updates must be **audit logged** with user and timestamp.
Special Requirements	• Support for MARC import/export. • Customizable cataloging templates for different materials (books, journals, videos, etc.).
Assumptions	• Catalogers are trained in standard cataloging procedures. • Library follows a defined classification and subject heading policy.
Notes & TBD	• Select external metadata sources for integration. • Finalize the complete list of mandatory fields per material type.
Related Misuse Cases	• Accidental deletion of critical records. • Entry of low-quality or inconsistent metadata.

Sequence Diagram for UC-LMS-12 Manage Library Cataloging

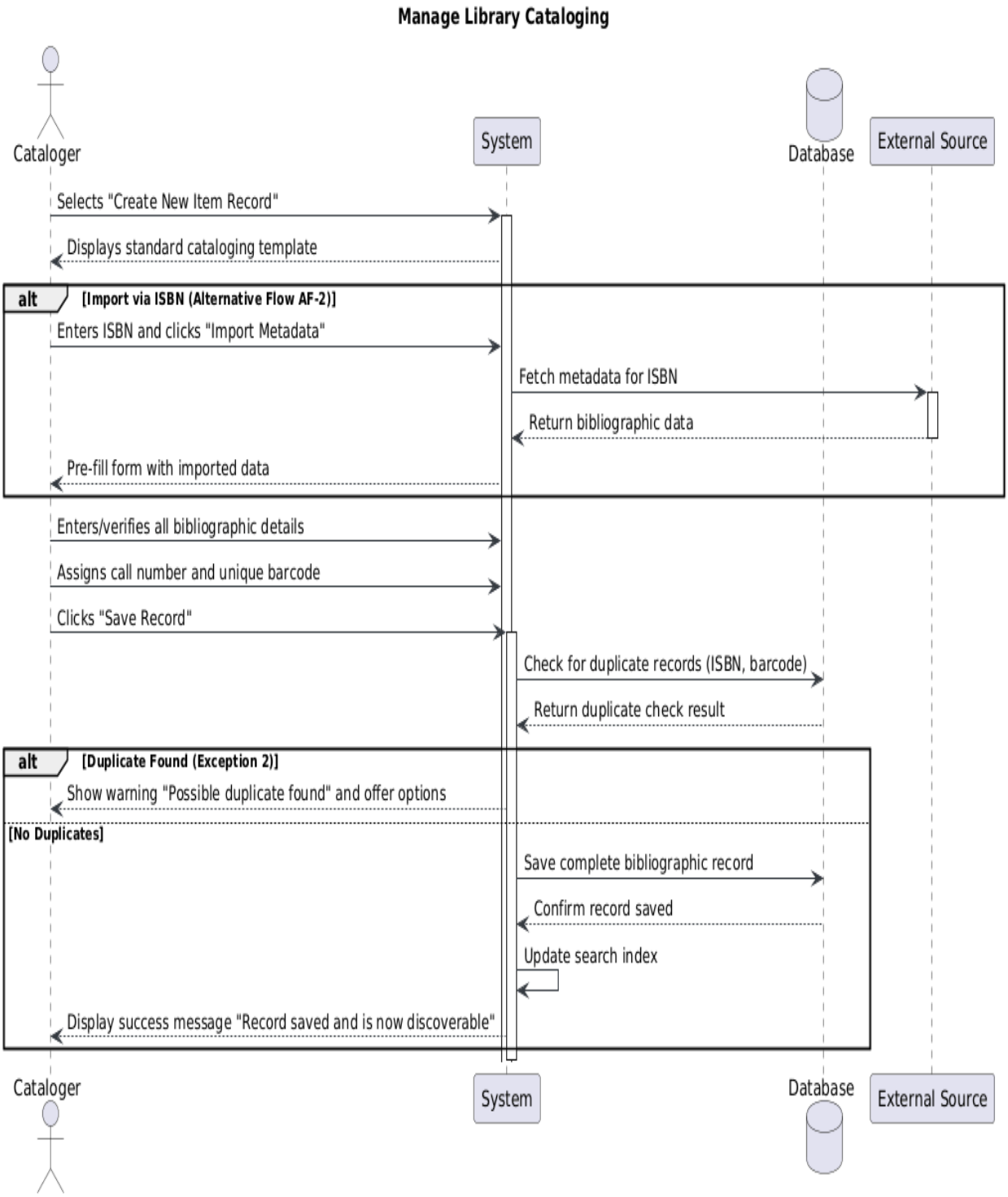

Figure 12 Sequence Diagram for UC-LMS-12 Manage Library Cataloging

Table 29 Manage Wikipedia Article Creation Use case

Field	Description
Use Case ID	UC-OWP-01
Use Case Name	Manage Wikipedia Article Creation
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Registered Editor
Description	Defines the steps taken by a Registered Editor to create a new encyclopedic article within the Oromo Wikipedia section of the platform.
Trigger	A Registered Editor notices a missing topic and decides to create a new article.
Preconditions	• User is authenticated as a **Registered Editor**. • Article title does **not** already exist in the wiki.
Normal Flow	1. Registered Editor navigates to the Oromo Wikipedia module. 2. They search for the intended article title; system displays: *"This page does not exist. You can create it."* 3. Registered Editor clicks **"create it"**. 4. System opens the rich-text wiki editor (supports markup, links, images). 5. Editor writes the article, adds citations, inserts media (free-license only). 6. Editor enters a brief summary in the **Edit Summary** section. 7. Editor clicks **Publish page** or **Submit for Review**. 8. System validates the content (spam filter, link validation). 9. System creates the page and saves its first version in revision history. 10. System either publishes immediately or sends the article to a **review/moderation queue** depending on policy.

Postconditions	• A new article is created in Oromo Wikipedia. • The edit is logged in **Recent Changes**. • If review is required, **Administrators/Sysops are notified**.
Alternative Flows	**AF-2.1 Article Title Already Exists:** • When searched, system finds an existing page and displays it. • Use case ends; user may continue with **UC-OWP-02: Edit Existing Article**. **AF-7.1 Save as Draft:** • Instead of publishing, user selects **Save Draft**. • System stores the draft in the user's private workspace.
Exceptions	**Exception 1: Edit Conflict** • Another user creates the same article simultaneously. • System detects conflict and shows an *Edit Conflict* resolution screen. **Exception 2: Abusive Content Detected** • Spam/profanity detected by filters. • System blocks the publish action, displays a warning, and flags the event for admin review. **Exception 3: System Errors** • Database or saving error occurs. • System logs the issue, shows a user-friendly error, and provides an option to copy or download work to prevent data loss.
Priority	High
Effort/Complexity	Medium
Frequency of Use	High (ongoing wiki growth)
Business Rules	• Article titles must be **unique**. • Uploaded media must have a **valid free license**.

	• Content must follow Wikipedia policies (Neutrality, Verifiability, No Original Research). • All edits must be logged with user ID and timestamp.
Special Requirements	• Editor must support both **wiki markup** and **visual WYSIWYG editing**. • System must maintain full **revision history** for version control.
Assumptions	• Registered Editors understand basic wiki editing and are contributing in good faith. • Community policies for article creation are already defined.
Notes and TBD	• Decide whether new articles from non-admins are auto-published or require moderation. • Finalize detailed rules for the automated spam/content filter.
Related Misuse Cases	• Creating pages for advertisement/self-promotion. • Repeated creation of offensive or nonsensical pages (vandalism).

Sequence Diagram for UC-OWP-01 Manage Wikipedia Article Creation

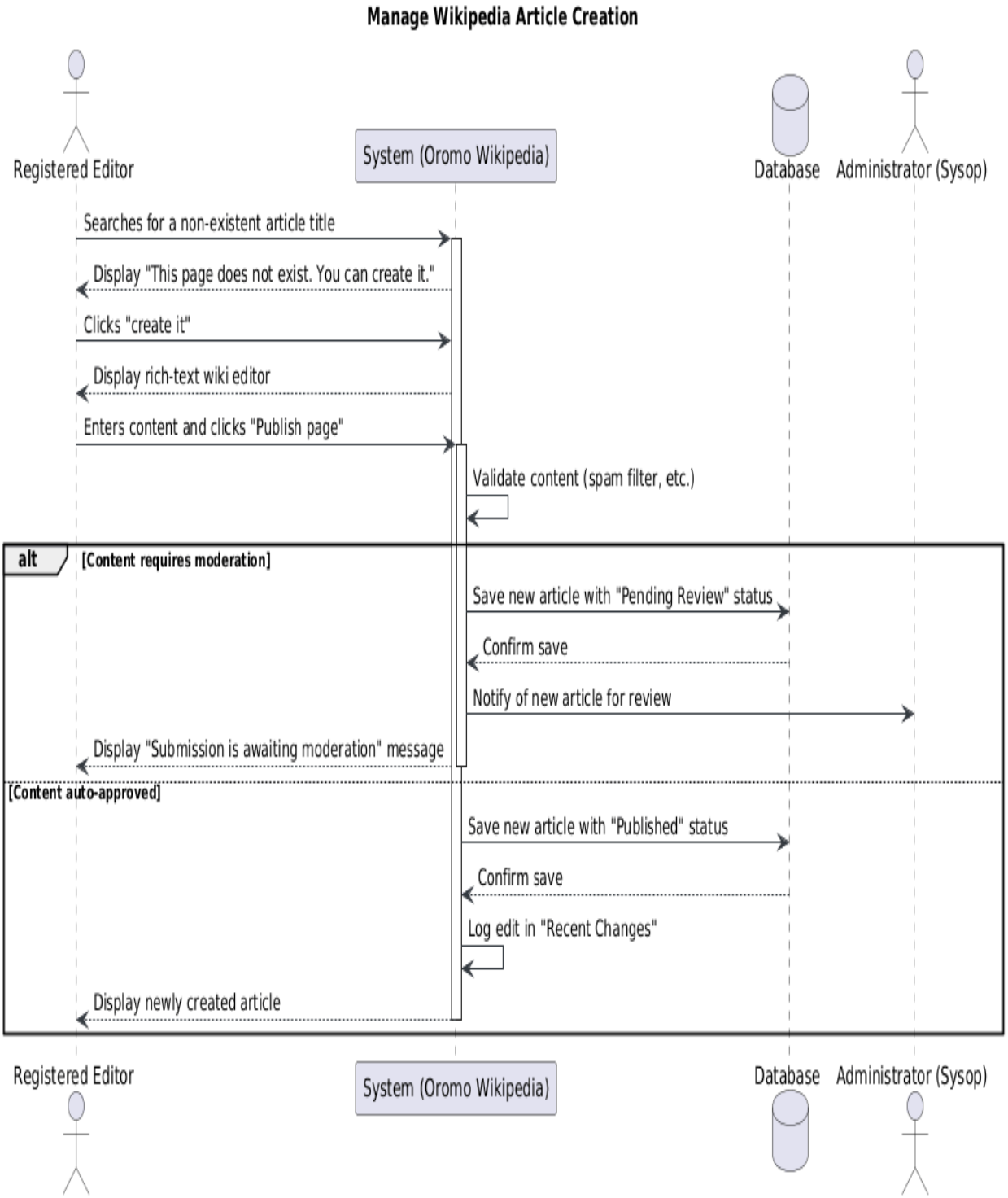

Figure 13 Sequence Diagram for UC-OWP-01 Manage Wikipedia Article Creation

Table 30 Manage Wikipedia Content Moderation Use case

Field	Description
Use Case ID	UC-OWP-03
Use Case Name	Manage Wikipedia Content Moderation
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Administrator (Sysop)
Description	This use case describes how a Wikipedia Administrator (Sysop) monitors, reverts vandalism, moderates content, enforces community policies, and manages user behavior to maintain the integrity of the Oromo Wikipedia.
Trigger	When the Sysop is monitoring recent changes or responding to reported disruptive behavior.
Preconditions	• The Sysop is authenticated and logged in. • The Sysop holds the "Administrator (Sysop)" role with elevated permissions.
Normal Flow (Reverting Vandalism & Blocking a User)	1. Sysop opens the "Recent Changes" log or watchlist. 2. Sysop identifies vandalism/disruptive edit (e.g., blanking a page). 3. Sysop clicks **Undo** next to the edit. 4. System restores the previous version and logs “Undid revision…” in history. 5. Sysop opens the user’s contributions page. 6. Sysop confirms repeated vandalism or malicious intent. 7. Sysop clicks **Block User**. 8. System displays a block configuration form (duration, reason, options). 9. Sysop selects duration, provides reason, and confirms block. 10. System enforces the block, updates logs, and prevents future edits by the user/IP.
Postconditions	• Vandalism is reverted. • User is blocked for the configured duration. • All actions are publicly logged for transparency.
Alternative Flows	**3.1 Page Deletion** 3.1.1 Sysop identifies a newly created policy-violating page. 3.1.2 Sysop clicks **Delete**. 3.1.3 Sysop provides deletion reason and confirms. 3.1.4 System removes page from public view and logs deletion. **3.2 Page Protection**

	3.2.1 Sysop identifies edit-warring or repeated disruptive edits. 3.2.2 Sysop opens the page and selects **Protect**. 3.2.3 Sysop sets protection level (e.g., Semi-protection) and duration. 3.2.4 System enforces protection and logs the action.
Exceptions	**Exception 1: Accidental Action** • Sysop mistakenly deletes a page or blocks the wrong user. • System must allow **Undelete** or **Unblock** actions by Sysops. • System logs all reversal actions as well. **Exception 2: System Error** • Database/server error during administrative action. • System must fail gracefully, log the error, and show a user-friendly message.
Priority	High
Effort/Complexity	High
Frequency of Use	Daily (continuous administrator responsibility)
Business Rules	• All admin actions must be logged with timestamp and username. • Blocked users must receive clear messages explaining the reason. • Deleted pages must remain recoverable by Sysops (soft deletion). • Admin decisions must follow community policies and governance protocols.
Special Requirements	• Efficient monitoring tools (watchlists, contribution logs, recent changes, filters). • Clear admin UI with confirmation prompts to prevent accidental actions. • Real-time updates of activities for quick response.
Assumptions	• Sysops understand wiki policies and act responsibly. • Community governance policies define what constitutes vandalism, policy violation, and allowed admin actions.
Notes & TBD	• Define user groups (e.g., “Established User,” “Autoconfirmed”). • Finalize standard block durations and predefined block reasons. • Define escalation matrix for repeated vandalism.
Related Misuse Case	• Administrator misuses power to influence content disputes (“wheel-warring”). • Compromised Sysop account causes mass deletions, mass blocks, or widespread vandalism.

Sequence Diagram for UC-OWP-03 Manage Wikipedia Content Moderation

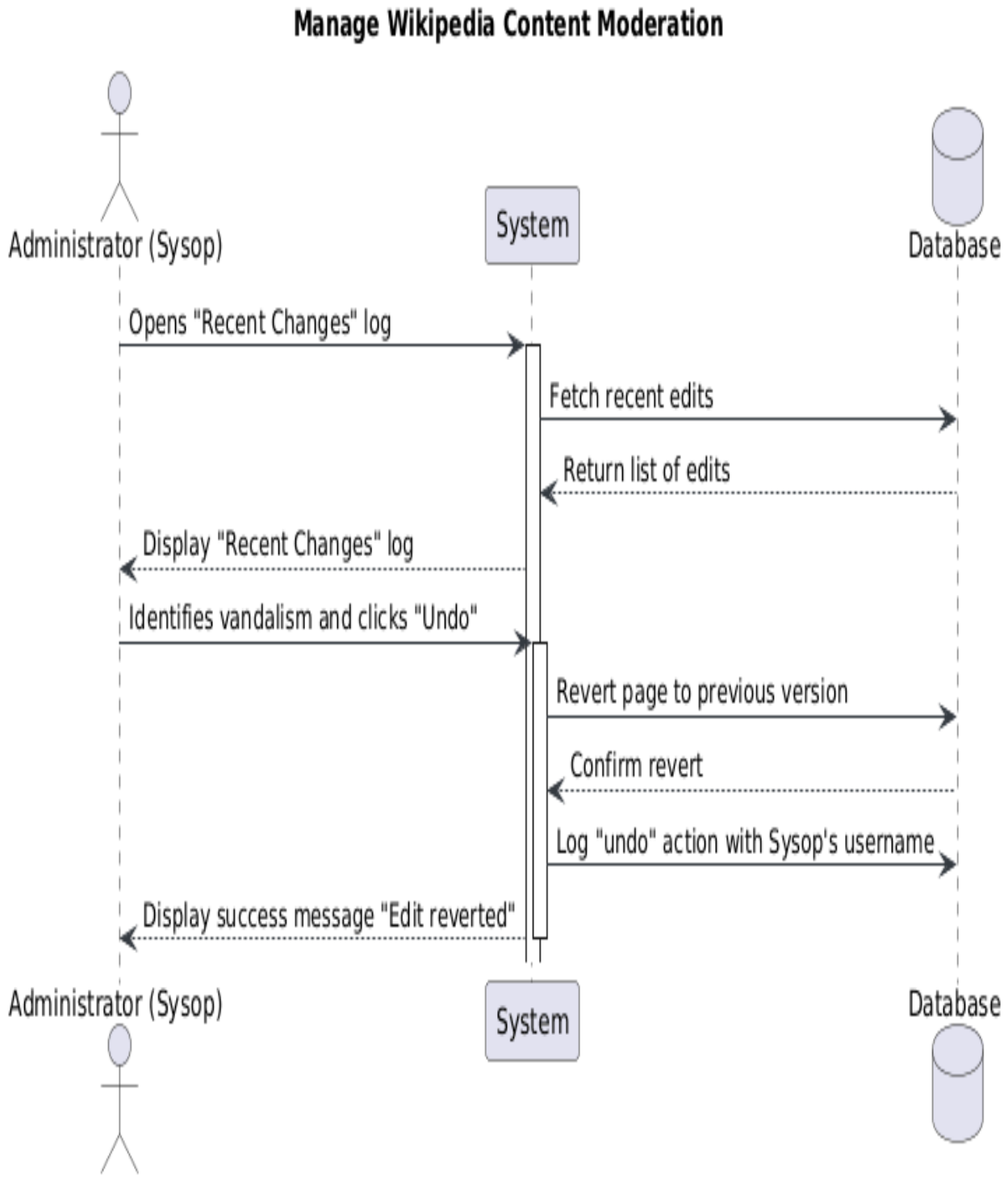

Figure 14 Sequence Diagram for UC-OWP-03 Manage Wikipedia Content Moderation

Table 31 Manage Repository Deposit Use case

Field	Description
Use Case ID	UC-RMS-01
Use Case Name	Manage Repository Deposit
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Researcher / Author
Description	This use case describes the process for a Researcher or Author to upload scholarly work (datasets, theses, pre-print articles, reports, etc.) and provide metadata for long-term preservation and access in the ORA Digital Repository.
Trigger	A researcher decides to archive or share their scholarly output in the Institutional Repository.
Preconditions	• User is authenticated and logged in with the "Researcher / Author" role. • User has the digital file(s) prepared for upload.
Normal Flow	1. User logs into the system and navigates to the **Repository** section. 2. User clicks **Start New Deposit / Upload Your Work**. 3. System loads the multi-step deposit wizard. 4. **Step 1 – File Upload:** User uploads primary file(s). 5. **Step 2 – Metadata Entry:** User fills in mandatory metadata (Title, Author(s), Publication Date, Resource Type). 6. User adds optional metadata (Abstract, Keywords, Subjects, References). 7. **Step 3 – Access & License:** User selects Access Level (Open, Restricted, Embargoed). 8. User chooses a License (e.g., Creative Commons). 9. **Step 4 – Review:** System displays a summary of files and metadata. 10. User verifies data, accepts the Deposit Agreement, and clicks **Complete Deposit**. 11. System validates required fields and file uploads. 12. System saves the deposit with *Pending Review* status and displays a confirmation. 13. System notifies the Repository Curator of the new submission.
Postconditions	• Deposit record and associated metadata/files are stored. • Item status is set to **Pending Review**. • Item is not yet visible to the public.
Alternative Flows	**5.1 Automatic Metadata Retrieval (DOI Lookup):** 5.1.1 User enters DOI and clicks **Fetch Metadata**.

	5.1.2 System retrieves metadata from an external service (e.g., Crossref). 5.1.3 User reviews and confirms auto-populated fields.
Exceptions	**Exception 1 – Missing Required Metadata** • User attempts to submit without mandatory fields. • System highlights missing fields and prevents submission. **Exception 2 – File Upload Failure** • File upload fails due to size, format, or network issue. • System shows an error message and allows retry *without losing metadata already entered*. **Exception 3 – System Error** • System encounters an error during data saving. • System logs the error, notifies the user, and attempts to save progress as a draft.
Priority	High
Effort/Complexity	High
Frequency of Use	High (core repository function; used daily by researchers)
Business Rules	• Each deposit must contain at least one file. • Mandatory metadata fields must follow repository standards (e.g., Dublin Core). • All depositors must accept the standard Deposit Agreement. • Metadata must comply with controlled vocabularies where applicable.
Special Requirements	• Support for diverse file formats (PDF, DOCX, ZIP datasets, images, video, etc.). • Inline help tooltips and metadata examples for users. • Support for large file uploads (e.g., datasets > 1GB, if infrastructure allows).
Assumptions	• Depositor has copyright or legal rights to upload and distribute the work. • User will provide accurate bibliographic and descriptive metadata.
Notes and TBD	• Finalize mandatory Dublin Core elements. • Define allowed access levels (Open, Restricted, Embargo) and CC license options. • Confirm maximum file size and upload limits.
Related Misuse Case	• User uploads copyrighted material without permission. • User uploads non-scholarly or inappropriate files.

Sequence Diagram for UC-RMS-01 Manage Repository Deposit

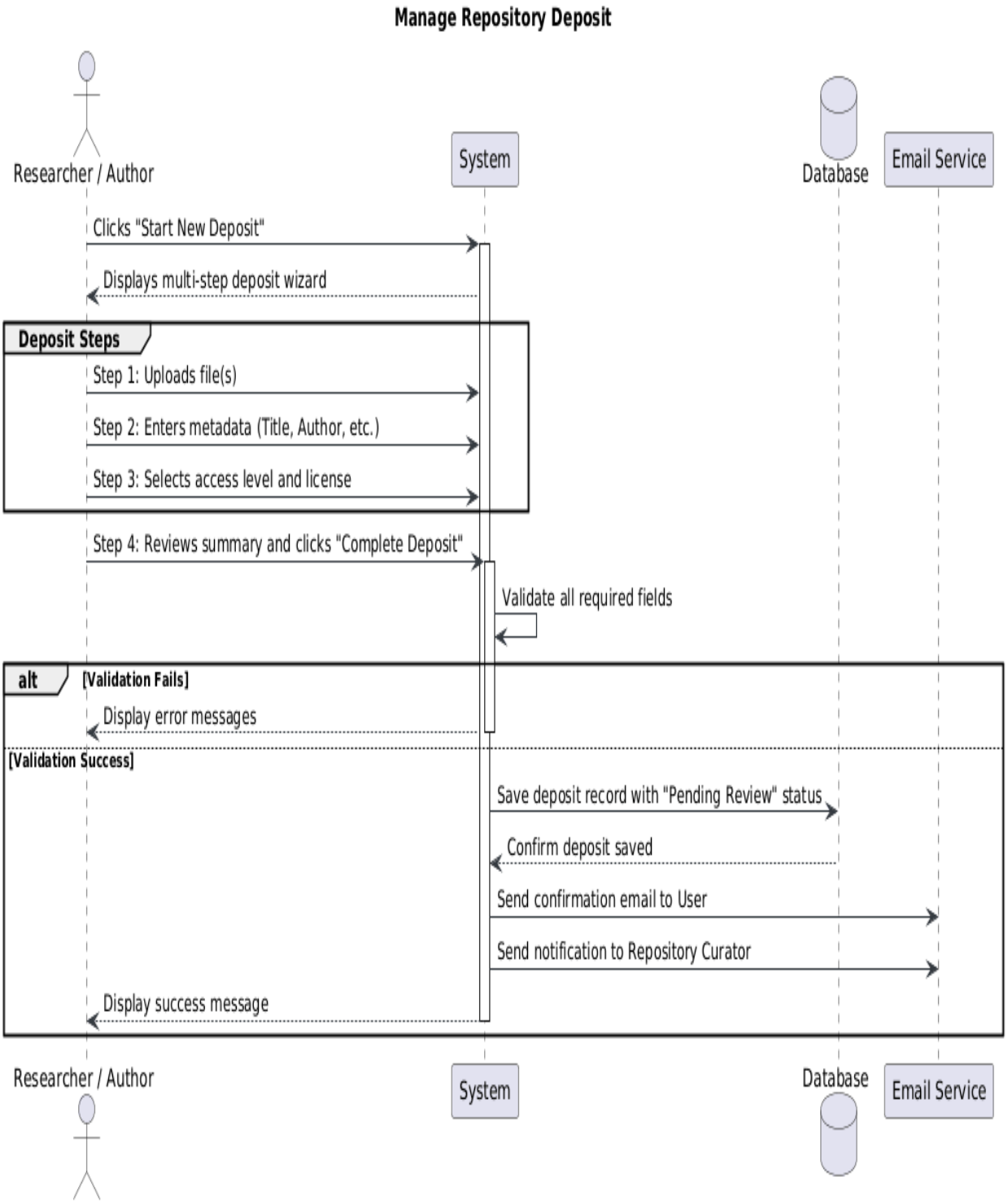

Figure 15 Sequence Diagram for UC-RMS-01 Manage Repository Deposit

Table 32 Manage Repository Curation Use case

Field	Description
Use Case ID	UC-RMS-16
Use Case Name	Manage Repository Curation
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Repository Curator
Description	This use case describes the curation workflow performed by the Repository Curator to review a new submission, validate and enhance its metadata, check policy compliance, and advance it to the next stage of approval in the repository lifecycle.
Trigger	The Curator receives a notification that a newly deposited item is awaiting review.
Preconditions	• Curator is authenticated with the **Repository Curator** role. • An item exists in **Pending Review** status.
Normal Flow	1. Curator logs into the system and opens the **Curation Dashboard**. 2. Curator selects an item listed under **Pending Review**. 3. System displays an admin-level view of the item, including files and metadata. 4. **Metadata Validation:** Curator inspects core bibliographic metadata (Title, Authors, Date, Resource Type) for accuracy, completeness, and conformance with standards (e.g., Dublin Core). 5. **Metadata Enrichment:** Curator enhances the record—adds subject terms from controlled vocabularies, standardizes author names, adds ORCID IDs, links related items, and improves descriptions. 6. **Policy Verification:** Curator verifies copyright compliance, license appropriateness, and access-level consistency (e.g., referencing Sherpa/RoMEO policies where relevant). 7. Curator clicks **Approve Metadata** after completing the review. 8. System saves all edits. 9. System updates the item's workflow status (e.g., *Pending Content Review* or *Pending Final Approval*) and notifies the next responsible role.
Postconditions	• Metadata is validated, corrected, enriched, and aligned with repository standards. • Item advances to the next stage in the publication workflow.
Alternative Flows	**7.1 Return Submission to Author** 7.1.1 Curator selects **Return to Author** if major problems exist.

	7.1.2 System prompts Curator to write a message explaining required changes. 7.1.3 System updates item status to **Returned**, notifies the Author, and allows resubmission. **7.2 Reject Submission** 7.2.1 Curator selects **Reject** if the deposit is entirely unsuitable (e.g., spam, irrelevant content). 7.2.2 Curator provides a reason. 7.2.3 System removes the item from the workflow, logs the action, and notifies the Author.
Exceptions	**Exception 1 – System Errors•** System fails to save metadata changes due to a database or technical error. • System logs the error, displays a message to the Curator, and ensures that no metadata edits are lost.
Priority	High
Effort/Complexity	High
Frequency of Use	Daily (critical repository function)
Business Rules	• Every submission must pass curation before being approved for publication. • Metadata must follow repository standards (e.g., Dublin Core, controlled vocabularies). • Curator must verify copyright status, chosen license, and access level. • All curation activity must be logged with username, timestamp, and action details.
Special Requirements	• System must provide advanced metadata editing tools (bulk editing, controlled vocabulary lookup, authority files). • System should support direct lookup or integration with copyright tools (e.g., Sherpa/RoMEO API). • Controlled vocabularies (e.g., LCSH, MeSH, AGROVOC) must be supported.
Assumptions	• Curator is trained in metadata standards, scholarly communication practices, and repository policies. • Repository workflow stages (Review → Curation → Approval → Publication) are well defined.
Notes and TBD	• Finalize list of controlled vocabularies to be used. • Clarify full multi-role workflow (e.g., is there a “Content Reviewer” or “Repository Administrator” after Curation?).
Related Misuse Cases	• Curator approves an item without verifying copyright or license. • Curator applies inconsistent metadata, harming repository data quality.

Sequence Diagram for UC-RMS-16 Manage Repository Curation

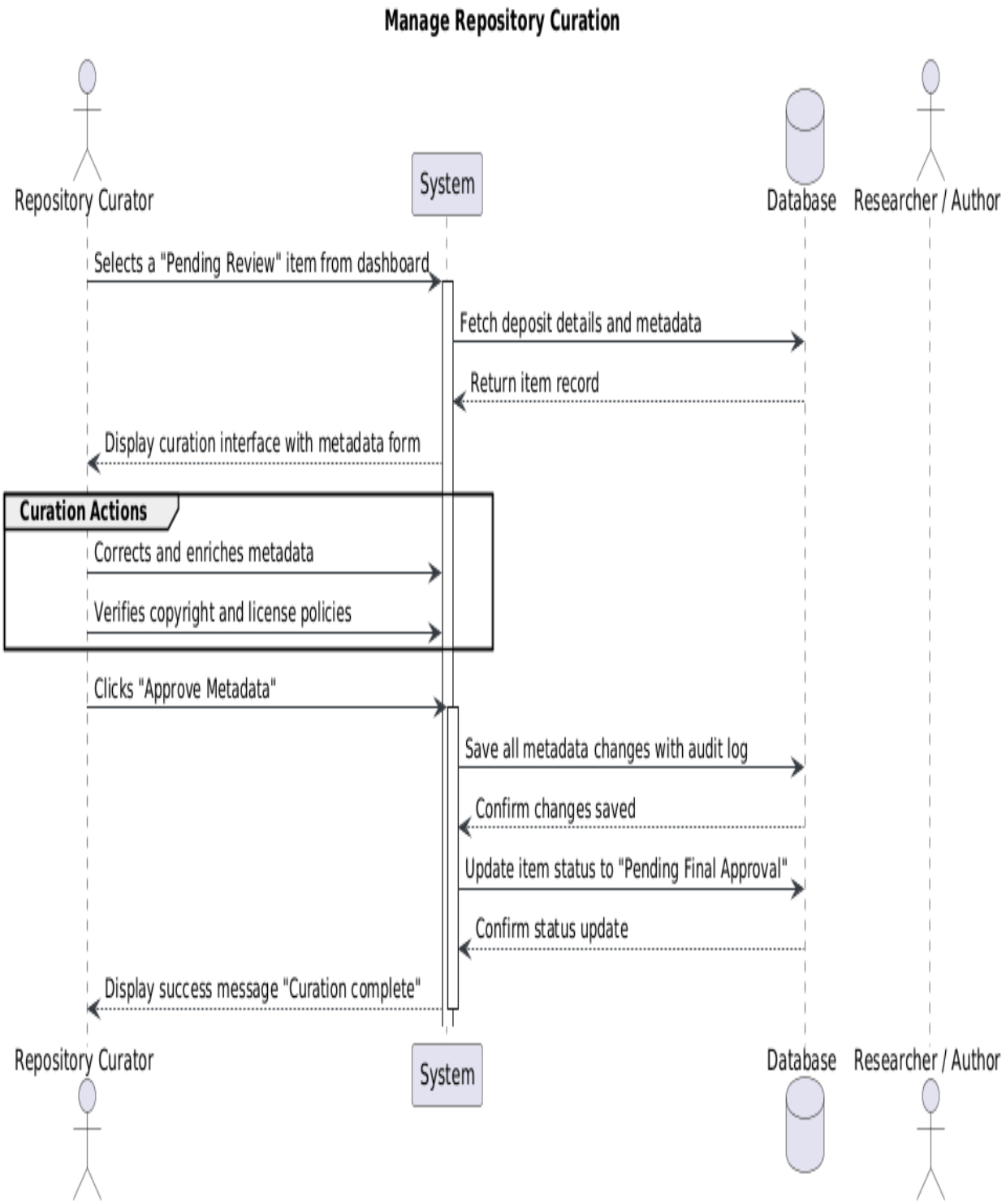

Figure 16 Sequence Diagram for UC-RMS-16 Manage Repository Curation

Table 33 Manage Researcher Profile Use case

Field	Description
Use Case ID	UC-RNP-01
Use Case Name	Manage Researcher Profile
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Researcher / Member
Description	This use case describes how a registered Researcher/Member creates, updates, and maintains their professional profile on the Researchers' Network Platform. The profile serves as the researcher's public academic identity.
Trigger	A new member sets up their profile after registration, OR an existing member updates their profile details.
Preconditions	• User is authenticated. • User holds the **Researcher / Member** role. • Basic registration data already exists.
Normal Flow – Creating & Updating Profile	1. Researcher logs into the system. 2. User navigates to **My Profile** from the dashboard. 3. System displays the current profile information (blank for new accounts). 4. User selects **Edit Profile**. 5. System opens a multi-section profile form. 6. User updates Personal Information (Name, Title, Biography). 7. User updates Academic Credentials (Degrees, Universities). 8. User updates Affiliations (Institutions, Departments). 9. User adds/updates Publications (Title, Year, Link/DOI). 10. User adds Research Interests (tags/keywords). 11. User uploads a profile picture. 12. User clicks **Save Changes**. 13. System validates all entered data. 14. System stores the updates and displays **"Profile Updated Successfully."**
Post Conditions	• The database contains an up-to-date version of the user's profile. • Updated information is immediately visible on the public profile page.
Alternative Flows	**AF-1: Linking Internal Publications** 7.1. System provides a search option for ORA repository/journals. 7.2. User selects their publication from search results.

	7.3. System auto-populates metadata and links it to the profile.
Exceptions	**Exception 1 – Validation Failure** • Invalid formats detected (e.g., malformed URL). • System shows inline error messages and blocks saving. **Exception 2 – File Upload Failure** • Unsupported format or file too large. • System displays error (e.g., "Only JPG/PNG under 5MB allowed"). **Exception 3 – System Error** • Database failure during save operation. • System logs the error and shows a friendly message without losing user input.
Priority	High
Effort / Complexity	Medium
Frequency of Use	Medium (frequent for new users, periodic for active researchers)
Business Rules	• Users can only edit **their own** profile. • Registered email may be non-editable. • Uploaded media must comply with platform guidelines. • Public profile fields are visible/searchable by all members (except private settings).
Special Requirements	• A Privacy Settings module allows users to control visibility of each section. • Integration with external networks (ORCID, Google Scholar, LinkedIn) should be supported.
Assumptions	• Users provide accurate, professional information.
Notes / TBD	• Finalize full profile field list. • Define default privacy settings values.
Related Misuse Cases	• Fake or impersonated profiles. • Uploading inappropriate profile photos. • Using profile fields for spam or advertising.

Sequence Diagram for UC-RNP-01 Manage Researcher Profile

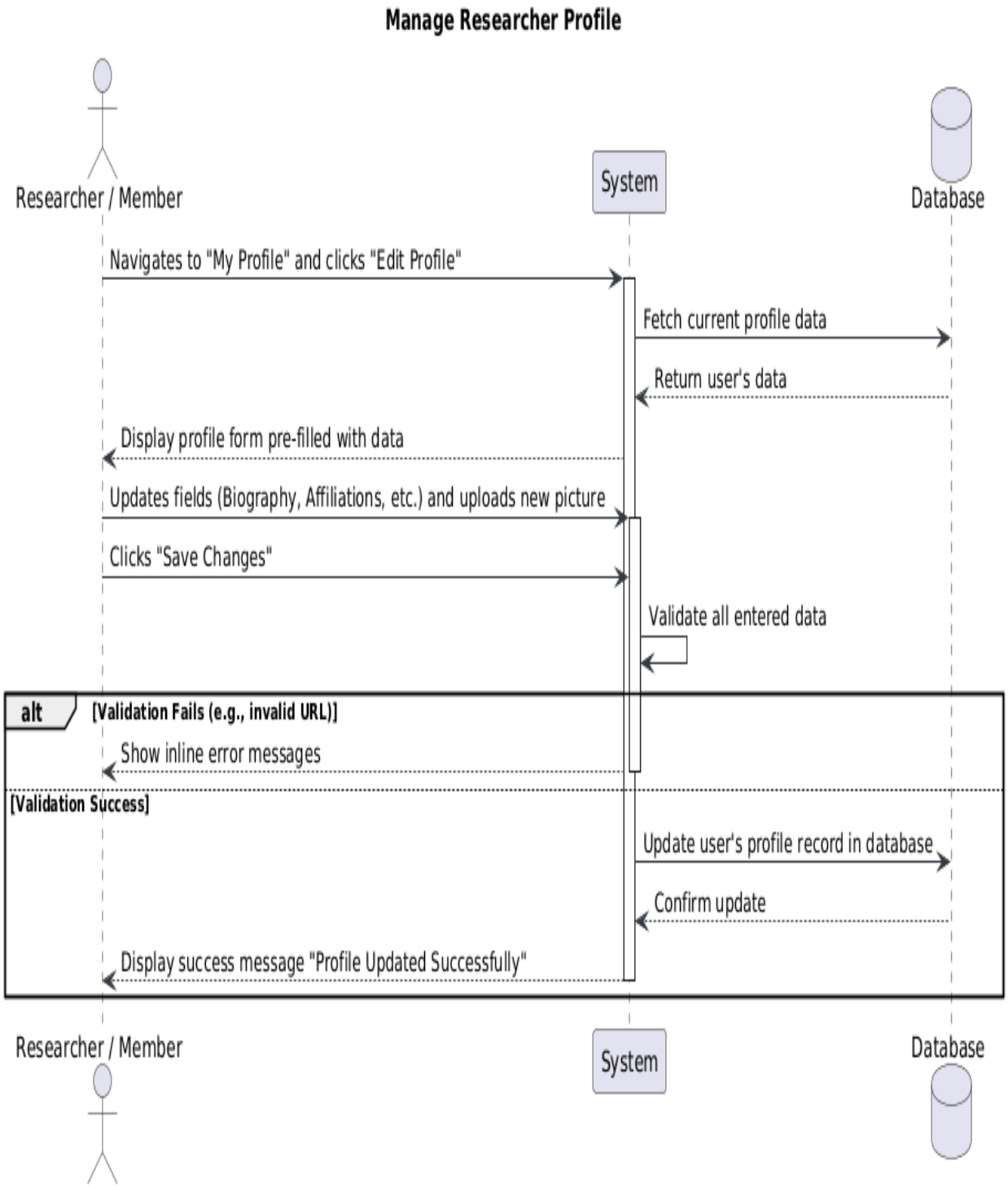

Figure 17 Sequence Diagram for UC-RNP-01 Manage Researcher Profile

Table 34 Manage Research Group Use case

Field	Description
Use Case ID	UC-RNP-18
Use Case Name	Manage Research Group
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Researcher / Member, Group Moderator
Description	This use case describes how a Researcher creates a new research group and how a Group Moderator manages membership, discussions, and group settings to ensure productive collaboration.
Trigger	A Researcher wants to create a new collaborative space, or a Group Moderator needs to perform administrative tasks within their group.
Preconditions	• User is authenticated. • User has the **Researcher / Member** or **Group Moderator** role. • Platform's "Research Groups" module is available.
Normal Flow – Creating a New Group	1. Researcher navigates to the **Research Groups** section. 2. Clicks **Create New Group**. 3. System displays the group setup form. 4. User enters Group Name, Description, and Keywords/Tags. 5. User sets Privacy Level (Public or Private). 6. User clicks **Create Group**. 7. System validates inputs, creates the group, and assigns the user as the first Group Moderator. 8. System navigates the user to the group's main page.
Post Conditions	• A new research group exists and is discoverable in the directory. • The creator is assigned as the Group Moderator with administrative privileges.
Alternative Flows	**AF-1: Managing Membership (Moderator)** 1. Private group: a member requests to join. 2. System notifies the Group Moderator. 3. Moderator reviews requests and clicks **Approve** or **Decline**. 4. System updates membership and notifies the requesting user. **AF-2: Moderating Discussions (Moderator)** 1. Moderator identifies posts/comments violating guidelines. 2. Moderator deletes the offending post. 3. If needed, Moderator removes the user from the group.
Exceptions	**Exception 1 – Duplicate Group Name**

	• Action: System prevents creation and displays: "A group with this name already exists. Please choose a different name." **Exception 2 – Unauthorized Action** • Action: Non-moderators attempting to access moderation tools are denied access. **Exception 3 – System Errors** • Action: System logs error, shows user-friendly message, and advises retrying the operation.
Priority	High
Effort / Complexity	High
Frequency of Use	Medium (group creation is occasional; moderation is frequent)
Business Rules	• Each group must have at least one Group Moderator. • Last moderator must assign a new moderator or the group may be archived. • Moderators manage only their assigned groups. • All members must follow platform community guidelines.
Special Requirements	• Each group has a discussion forum, member list, and shared document area. • System sends timely notifications for membership requests and their status updates.
Assumptions	• Users create groups for legitimate academic collaboration. • Moderators act responsibly to maintain a productive environment.
Notes / TBD	• Define full moderator toolset (e.g., temporary bans, pin posts). • Define process for reporting groups or moderators for misconduct.
Related Misuse Cases	• Users create spam or fraudulent groups. • Moderators unfairly remove members. • Group discussions are used for harassment or misinformation.

Sequence Diagram for UC-RNP-02 Manage Research Group

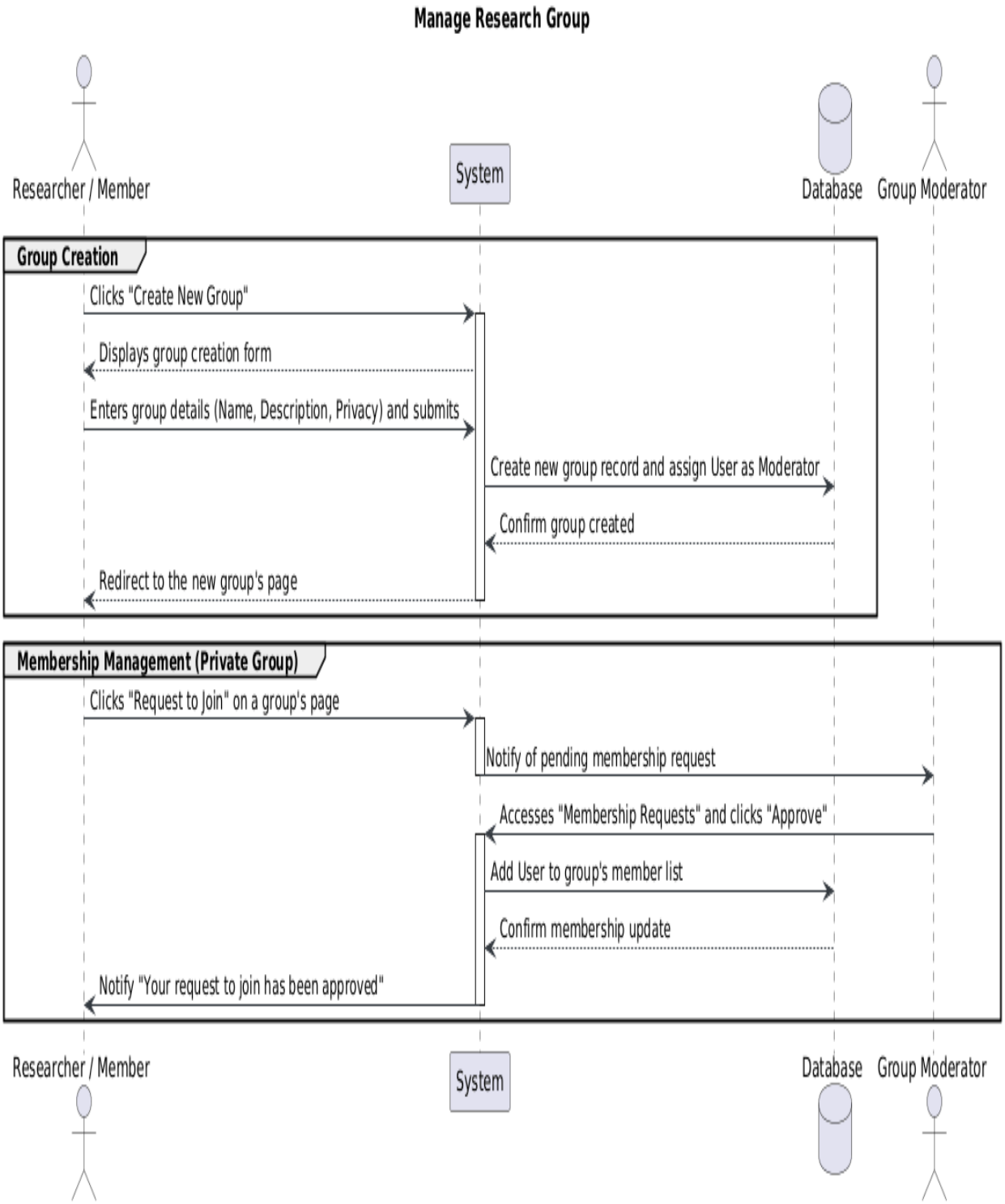

Figure 18 Sequence Diagram for UC-RNP-02 Manage Research Group

Table 35 Manage Citation Generation Use case

Field	Description
Use Case ID	UC-CITE-01
Use Case Name	Manage Citation Generation
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Any User (Public User, Researcher / Member, etc.)
Description	This use case describes how a user can generate and copy a pre-formatted, accurate citation for any citable item on the ORA Digital Platform.
Trigger	A user is viewing a piece of content (e.g., journal article, eBook, repository item) and wants to cite it.
Preconditions	• User is on the item's page. • The item has complete metadata (author, title, date, DOI, etc.) stored in the system.
Normal Flow	1. User clicks the **Cite** button on the item's page. 2. System displays a pop-up window with the citation tool. 3. System generates and shows the citation in multiple styles (e.g., APA, MLA, Chicago). 4. User selects the desired citation style. 5. User clicks **Copy** or **Copy to Clipboard** next to the selected style. 6. System copies the citation to the user's clipboard. 7. System shows a confirmation message (e.g., "Citation copied!"). 8. User closes the pop-up and can paste the citation into their document.
Post Conditions	• Citation is successfully copied to the user's clipboard.
Alternative Flows	• Exporting citations to reference management tools is covered in **UC-CITE-02**.
Exceptions	**Exception 1 – Incomplete Metadata** • System generates the citation with placeholders (e.g., "[Year Missing]") and shows a warning message. **Exception 2 – System Errors** • Client-side errors prevent pop-up or copy functionality. • System should fail gracefully, display citations inline for manual copy, and log the error for troubleshooting.
Priority	High
Effort / Complexity	Medium
Frequency of Use	Very High (core feature for proper citation of platform content)
Business Rules	• Citations must accurately reflect stored metadata.

	• Items with a DOI must include the full DOI link. • The system must support at least APA, MLA, and Chicago styles. • Citation tool must be accessible to all users, logged in or not.
Special Requirements	• Citation formatting must strictly follow official style guides. • **Copy to Clipboard** must work reliably across supported browsers.
Assumptions	• Users know which citation style they need. • Metadata in the ORA database is accurate and well-maintained.
Notes / TBD	• Finalize full list of supported citation styles. • Plan updates for new editions of style guides (e.g., APA 7, MLA 9).
Related Misuse Cases	• User copies citation without verifying accuracy, leading to incorrect references.

Sequence Diagram for UC-RNP-03 Manage Citation Generation

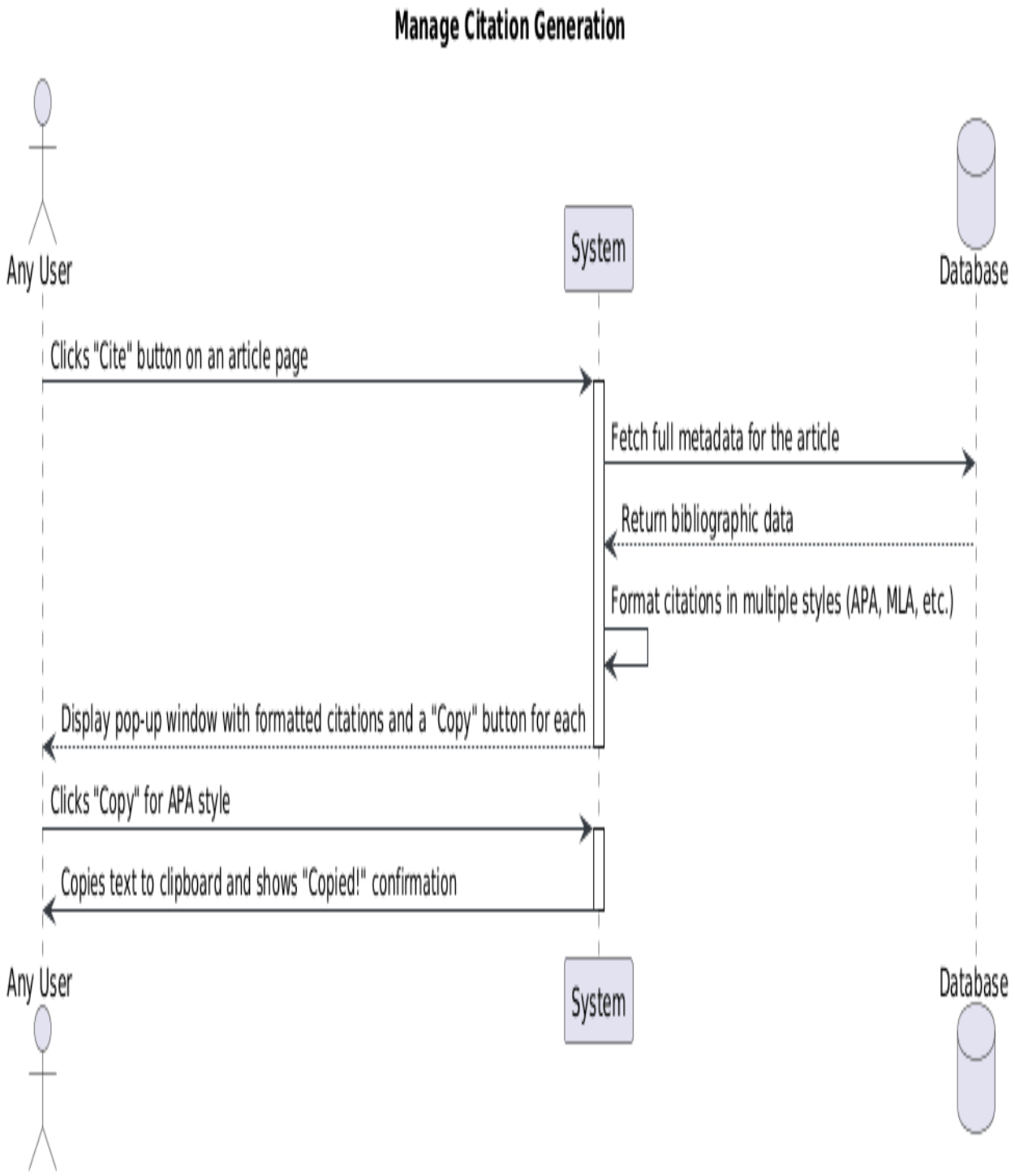

Figure 19 Sequence Diagram for UC-RNP-03 Manage Citation Generation

Table 36 Manage Citation Export Use case

Field	Description
Use Case ID	UC-CITE-02
Use Case Name	Manage Citation Export
Created By	Analysis and Design Team
Date Created	[Date]
Actors	Any User (Public User, Researcher / Member, etc.)
Description	This use case describes how a user can download bibliographic metadata of a citable item in a standardized format for direct import into reference management software (e.g., Zotero, Mendeley, EndNote).
Trigger	A user is viewing a citable item and wants to add it to their personal reference library without manually entering metadata.
Preconditions	• User is on the item's page. • The item has complete metadata stored in the system.
Normal Flow	1. User clicks the **Cite** button on the item's page. 2. System displays a pop-up window with the citation tool. 3. User selects the **Export Citation** or **Download for Reference Manager** section. 4. User clicks the desired export format link (e.g., **RIS (.ris)** or **BibTeX (.bib)**). 5. System generates a plain-text file containing the item's metadata in the selected format. 6. Browser initiates download of the generated file (e.g., citation.ris). 7. User opens the file in their reference management software, which imports all metadata automatically.
Post Conditions	• User has successfully downloaded a standardized file containing complete metadata for the item.
Alternative Flows	N/A (The user either downloads the file in the selected format or does not.)
Exceptions	**Exception 1 – Incomplete Metadata** • System generates the export file with available data. Missing fields are omitted according to format rules. **Exception 2 – System Errors** • Server-side error prevents file generation. • System logs the error and shows a user-friendly message, e.g., "Unable to generate export file at this time. Please try again later."
Priority	High

Effort / Complexity	Medium
Frequency of Use	High (especially among active researchers and students)
Business Rules	• System must generate valid, well-formed files for each supported format (RIS, BibTeX). • Metadata fields must correctly map to standard tags (e.g., PY - in RIS; year = {} in BibTeX). • DOI must always be included for items that have one. • File names should be descriptive or based on the item title (e.g., Author2025_ArticleTitle.ris).
Special Requirements	• File generation must be instant without page reload. • System must be compatible with major reference management software (Zotero, Mendeley, EndNote).
Assumptions	• User has reference management software installed that supports the downloaded file type. • ORA database metadata is accurate and up-to-date.
Notes / TBD	• Determine if additional export formats are needed (e.g., EndNote XML). • Define a standard naming convention for exported files.
Related Misuse Cases	• A user may attempt repeated downloads to put load on the server (low risk).

Sequence Diagram for UC-RNP-04 Manage Citation Export

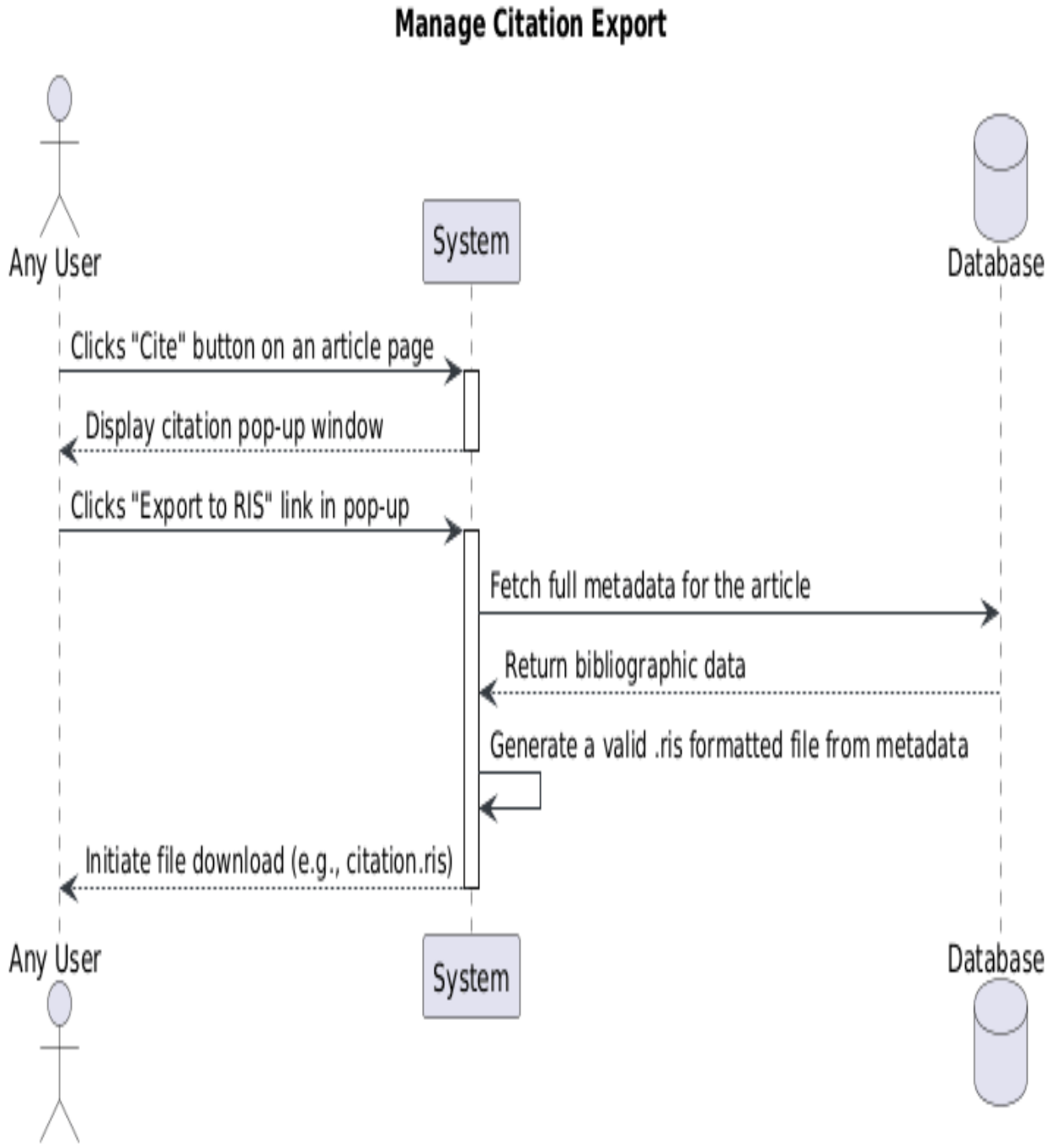

Figure 20 Sequence Diagram for UC-RNP-04 Manage Citation Export

5 System Architecture & Data Requirements

This section describes the high-level technical architecture, data management policies, and metadata standards for the ORA Digital Platform.

5.1 System Architecture Overview

The ORA Digital Platform adopts a cloud-based, modular, and service-oriented architecture designed for scalability, maintainability, and high availability. The architecture separates concerns into logical tiers, allowing for independent development, deployment, and scaling of components.

5.1.1 Architectural Style

- **Multi-tier Architecture:** The system is structured into distinct logical layers (Presentation, Application, Data) to isolate business logic from the user interface and data storage.
- **Modular Design:** The platform is composed of distinct modules (e.g., Journal Management, Library Management) that are designed to be loosely coupled. While they are integrated into a cohesive platform, they can be developed and updated independently.

5.1.2 High-Level Architecture Components

- **Presentation Layer (Frontend):** A responsive single-page application (SPA) built with **React.js**. It is responsible for rendering the user interface and managing all client-side interactions.
- **Application Layer (Backend):** A set of services built with **Node.js (Express)** or **Spring Boot (Java)**. This layer contains all the business logic, handles API requests from the frontend, manages user authentication, and orchestrates communication with the data layer and external services.
- **Data Layer:** This layer is responsible for data persistence and storage. It comprises:
    - **PostgreSQL Database:** A relational database managed via **AWS RDS** for storing all structured data, such as user information, metadata for publications, and workflow statuses.
    - **Object Storage (AWS S3):** A scalable storage service for all unstructured data, including uploaded manuscript files, generated PDFs, images, and other digital assets.
- **Integration Layer:** This layer manages communication with all third-party services. Key components include API clients for the **DOI Registration Agency (Crossref)**, **ISBN registry services**, and the designated **Payment Gateway**.

5.1.3 Deployment Architecture (AWS-Based)

The system will be deployed on Amazon Web Services (AWS) for scalability and reliability.

- **Compute:** The backend application will be containerized using **Docker** and deployed on **Amazon EC2** instances or managed via **Amazon ECS (Elastic Container Service)** for automated scaling and management.
- **Database:** A managed **PostgreSQL** instance will be provisioned using **Amazon RDS (Relational Database Service)** to handle automated backups, patching, and scaling.
- **Storage:** All user-uploaded files and digital assets will be stored in **Amazon S3 (Simple Storage Service)** for durability and scalability.
- **Monitoring:** System health, performance, and logs will be monitored using **Amazon CloudWatch**.

5.2 System Context Diagram

(A formal diagram would be inserted here, but the description is as follows)
The ORA Digital Platform operates as the central system. It provides services to its **Actors** (e.g., Public User, Researcher, Administrator). It interacts with several key **External Systems**:

- It sends requests to and receives data from the **DOI Registration Agency**.
- It securely communicates with a **Payment Gateway** to process transactions.
- It sends notifications via an **Email Service Provider**.
- It may query external **Bibliographic Databases** for metadata import.

5.3 Data Requirements

5.3.1 Database Requirements

- **Normalization**: The database schema shall be normalized to at least the Third Normal Form (3NF) to ensure data integrity and reduce redundancy.
- **Audit Timestamps:** All primary tables shall include created_at and updated_at timestamps to automatically track when a record is created and last modified.
- **Data Integrity:** Foreign key constraints, checks, and transactions shall be used to ensure the referential integrity and consistency of data across the database.

5.3.2 Data Retention

- **Published Scholarly Works:** All published journal articles, eBooks, and open-access repository items, along with their core metadata, shall be retained **permanently**.

- **User Accounts:** User accounts will be retained indefinitely unless a user requests deletion. Inactive accounts may be disabled after a defined period (e.g., 2 years).
- **System and Audit Logs:** Application and security logs shall be retained for a period of **one (1) year** for troubleshooting and auditing purposes.

5.4 Metadata Requirements Per Module

The system shall enforce the following minimum metadata standards:

- **Journal Article Metadata:** Title, Author(s) with affiliations, Abstract, Keywords, Journal Title, Volume, Issue, Publication Date, DOI, ISSN.
- **Book Metadata:** Title, Author(s)/Editor(s), Abstract/Synopsis, ISBN, DOI, Publication Date, Publisher.
- **Repository Metadata (Dublin Core):** The system shall implement the 15 core elements of the Dublin Core Metadata Standard, including Title, Creator, Subject, Description, Publisher, Date, and Identifier.

6 Api Requirements

If the platform exposes an API for external use, it will adhere to the following standards.

6.1 API Style Guide

All API endpoints will follow RESTful principles, use JSON format for requests and responses, and use camelCase for attribute names.

6.2 Authentication API Requirements

Access to protected API endpoints shall be secured using **JWT (JSON Web Tokens)**

6.3 Versioning

The API will be versioned in the URL path (e.g., /api/v1/).

6.4 Rate Limiting

Rate limiting will be implemented to prevent abuse (e.g., 100 requests/minute for public endpoints).

7 Acceptance Criteria

7.1 General Acceptance Criteria

The ORA Digital Platform project will be considered complete and accepted when:

1. All Functional Requirements (FRs) and Non-Functional Requirements (NFRs) defined in this SRS document have been implemented, tested, and verified.
2. The system passes a mandatory, independent third-party security audit.
3. All required user and technical documentation has been delivered.
4. ORA administrators and staff have received comprehensive training.

7.2 Module-Based Acceptance Criteria

This section defines the high-level acceptance criteria for each major module. A module will be considered complete and accepted only when all its corresponding criteria are met and verified.

1. Public Landing Page

- **Accepted when:** A first-time visitor can successfully navigate to the main URL, clearly understand the mission of the Oromo Research Association, and easily find and click direct links to all six primary modules. Dynamic content sections must be present and showcasing recent activity from the platform.

2. Authentication & Core System

- **Accepted when:** A new **Public User** can complete the entire registration workflow, including receiving and clicking a verification email to activate their account. Subsequently, a registered user must be able to securely log in to receive a persistent session and log out to terminate it. The "Forgot Password" workflow must be fully functional.

3. Journal Management System (JMS)

- **Accepted when:** An **Author** can successfully submit a manuscript from start to finish. The submission must then be successfully processed through the entire double-blind peer-review workflow: an **Associate Editor** can assign **Reviewers**, Reviewers can submit their feedback, and the **Editor-in-Chief** can make a final, binding decision. Finally, an accepted article must be successfully processed by a **Journal Manager** through copyediting and production, culminating in its publication on the platform with an officially registered **DOI**.

4. eBook Publishing System (BPS)

- **Accepted when:** An **Author** can submit a book-length manuscript, a **Book Editor** can manage its peer-review process and accept it, a **Finance & Operations Officer** can provide financial clearance, and a **Digital Content Manager** can successfully convert it into final PDF/EPUB formats, assign an official **ISBN** and **DOI**, and publish it to the platform's library.

5. Library Management System (LMS)

- **Accepted when:** A **Member** can successfully search the catalog and find both a physical and a digital book. The user must be able to view the real-time availability of the physical book and successfully place a hold on it if it is checked out. A **Librarian** must be able to use a barcode scanner to successfully complete the check-out and check-in process for a physical book. A **Cataloger** must be able to create a new, fully compliant bibliographic record for a new book and have it appear correctly in public search results.

6. Oromo Wikipedia Platform (OWP)

- **Accepted when:** A **Registered Editor** can successfully create a new article and edit an existing one, with all changes correctly tracked in the page's revision history. An **Administrator (Sysop)** must be able to successfully perform core moderation actions: revert a vandalistic edit, delete a policy-violating page, and temporarily block a disruptive user from editing.

7. Repository Management System (RMS)

- **Accepted when:** A **Researcher** can successfully deposit a new item (e.g., a dataset) by completing the multi-step deposit wizard, including uploading a file, providing Dublin Core metadata, and selecting an access license. The submitted item must then appear in a **Repository Curator's** dashboard, who must be able to review, edit, and approve the submission to advance it in the workflow.

8. Researchers' Network Platform (RNP)

- **Accepted when:** A new **Researcher / Member** can successfully create and fully customize their professional profile. The user must be able to search for another member by name or research interest and establish a connection. A member must also be able to successfully create a new research group (public or private), and as a **Group Moderator**, be able to approve a pending membership request for that group.

9. Citation Generation Tools

- **Accepted when:** Any **User**, on viewing a citable item (article, book, or repository item), can successfully generate an accurate, pre-formatted citation in all supported styles (APA, MLA, etc.). The user must also be able to successfully export the citation data in

both **RIS** and **BibTeX** formats and verify that the downloaded file can be correctly imported into a standard reference manager (e.g., Zotero).

8 Requirement Traceability Matrix (RTM)

This section provides a traceability matrix to map system requirements to their corresponding use cases and verification test cases. This matrix ensures that every requirement is addressed in the system design and has a corresponding method for verification. The "Test Case ID" is a placeholder that will be filled in by the Quality Assurance (QA) team.

Table 37 Requirement Traceability Matrix (RTM)

Requirement ID	Requirement Description	Use Case ID	Module	Test Case ID
Authentication & Core System				
FR-UM-001	System shall support self-registration for Public Users.	UC-SYS-01	User Management	TC-AUTH-001
FR-UM-002	System Administrators shall have the ability to create, modify, suspend, or deactivate user accounts.	UC-SYS-04	User Management	TC-ADMIN-001
FR-UM-004	System shall provide a secure “Forgot Password” workflow including email verification.	UC-SYS-03	User Management	TC-AUTH-002
Journal Management System (JMS)				
FR-JMS-001	An Author shall be able to submit a manuscript with required metadata and files.	UC-JMS-01	Journal Management	TC-JMS-001
FR-JMS-003	A Reviewer shall be able to submit comments, attachments, and a recommendation.	UC-JMS-04	Journal Management	TC-JMS-004
FR-JMS-007	System shall automatically generate and assign a unique DOI upon article publication.	UC-JMS-10	Journal Management	TC-JMS-010

eBook Publishing System (BPS)				
FR-BPS-001	A Digital Content Manager shall be able to upload eBook files and register metadata (ISBN, DOI, Contributors).	UC-BPS-09	eBook Publishing	TC-BPS-009
FR-BPS-003	System shall support monetized access to eBooks, managed by a Finance & Operations Officer.	UC-BPS-05	eBook Publishing	TC-BPS-005
Library Management System (LMS)				
FR-LM-001	A Digital Librarian and Cataloger shall be able to catalog both physical and digital books.	UC-LMS-12	Library Management	TC-LMS-012
FR-LM-005	A Member / User / Student shall be able to place a hold or reservation on a library item.	UC-LMS-04	Library Management	TC-LMS-004
FR-LM-008	System shall support barcode scanning for item identification and circulation operations.	UC-LMS-02	Library Management	TC-LMS-002
Oromo Wikipedia Platform (OWP)				
FR-OWP-001	A Registered Editor shall be able to create new encyclopedia articles.	UC-OWP-01	Oromo Wikipedia	TC-OWP-001

FR-OWP-005	System shall implement a role-based moderation workflow where a Sysop must approve major edits.	UC-OWP-03	Oromo Wikipedia	TC-OWP-003
Repository Management System (RMS)				
FR-RM-001	A Researcher / Author shall be able to upload various scholarly documents (papers, theses, datasets).	UC-RMS-01	Repository Management	TC-RMS-001
FR-RM-002	System shall require Dublin Core metadata for all deposited items.	UC-RMS-01	Repository Management	TC-RMS-001a
Researchers' Network Platform (RNP)				
FR-RN-001	A Researcher / Member shall be able to create and customize a personal profile.	UC-RNP-01	Researchers' Network	TC-RNP-001
FR-RN-004	Platform shall support creation of public or private discussion forums and research groups.	UC-RNP-02	Researchers' Network	TC-RNP-002
Citation Tools				
FR-CITE-001	System shall provide a "Cite" button that generates citations in standard formats.	UC-CITE-01	Citation Tools	TC-CITE-001
FR-CITE-005	System shall support export in RIS (.ris) and BibTeX (.bib) formats.	UC-CITE-02	Citation Tools	TC-CITE-002

Non-Functional Requirements (NFR)				
NFR-SEC-001	All user passwords must be protected using a strong, one-way salted hashing algorithm.	UC-SYS-01	System-Wide (Security)	TC-SEC-001
NFR-PERF-001	Average page load time for public pages must be under 3 seconds under normal load.	UC-LMS-01, UC-SYS-05	System-Wide (Performance)	TC-PERF-001

9 Error Handling & Logging Requirements

This section defines the requirements for handling errors, logging system events, and monitoring the health of the ORA Digital Platform. A consistent and robust strategy for error handling and logging is essential for system stability, security, and maintainability.

9.1 Error Handling

All error messages displayed to end-users shall be clear, concise, and constructive.

- **NFR-EH-001 (Clarity and Simplicity):** Error messages must be written in plain language (e.g., English, Afan Oromo) and avoid technical jargon. For example, display "Invalid email or password" instead of "Authentication Error #5001."
- **NFR-EH-002 (Constructive Guidance):** Whenever possible, error messages shall guide the user toward a solution. For example, "Password must be at least 8 characters long and contain one special character" instead of just "Invalid password."
- **NFR-EH-003 (Graceful Failure):** The system must handle unexpected errors gracefully without crashing or exposing sensitive information. A generic, friendly message shall be displayed for critical, unrecoverable server errors (e.g., "The system is temporarily unavailable. Please try again later."), while the technical details are logged internally.
- **NFR-EH-004 (Inline Validation):** For all data entry forms, the system should provide real-time, inline validation where possible, displaying an error message next to the invalid field before the user attempts to submit the form.
- **NFR-EH-005 (Preservation of Input):** When a form submission fails due to validation errors, the system must return the user to the form with their previously entered (valid) data intact, so they only need to correct the invalid fields.

9.2 Logging Requirements

The system shall implement a comprehensive logging strategy with different levels and purposes to support auditing, debugging, and security monitoring.

- **NFR-LOG-001 (Logging Levels):** The backend application shall generate logs at different severity levels. At a minimum, it must support:
    - **INFO:** For routine events, such as a user logging in or a service starting.
    - **WARN:** For unexpected but non-critical issues, such as a retryable connection to an external service.

- **ERROR:** For critical errors that prevent a function from completing, such as a database connection failure or an unhandled exception.

- **NFR-LOG-002 (Structured Logging):** All logs shall be written in a structured format (e.g., **JSON**). Each log entry must include, at a minimum: a precise **timestamp**, the **severity level**, a **correlation ID** to trace a single request across the system, and a descriptive **message**.
- **NFR-LOG-003 (Critical Action Logging):** The system shall maintain a separate, immutable audit log to record all security-sensitive and critical business actions. This log is for auditing and not for debugging. Actions to be logged include:
  - User login (successful and failed attempts).
  - User account creation, modification, or deletion.
  - Changes in user permissions or roles.
  - Final publication of a journal article or eBook.
  - Permanent deletion of any content.
  - Changes to system-wide settings.
- **NFR-LOG-004 (Audit Log Content):** Each audit log entry must contain the **timestamp**, the **actor** (which user performed the action), the **action** performed, the **target resource** (e.g., "Article ID: 123"), and the source **IP address** of the request.

9.3 Monitoring

- **NFR-MON-001 (Health Checks):** The system shall expose a health check endpoint (e.g., /api/health) that can be used by monitoring services to verify that the application and its database connection are operational.
- **NFR-MON-002 (Performance Monitoring):** Key performance metrics, such as API response times, CPU utilization, and database query performance, shall be collected and visualized in a monitoring dashboard (e.g., using Amazon CloudWatch).
- **NFR-MON-003 (Alerting):** An automated alerting mechanism shall be configured to notify the **System Administrator** via email or another channel in the event of critical system errors, such as a sustained spike in server errors or unreachability of the application.

10 Backup, Disaster Recovery & Maintenance Plan

This section outlines the requirements and plans for data backup, disaster recovery (DR), and routine system maintenance. These policies are critical for ensuring data integrity, business continuity, and the long-term health of the ORA Digital Platform.

10.1 Backup Requirements

A comprehensive and automated backup strategy shall be implemented to protect all critical system data from loss or corruption.

- **NFR-BCK-001 (Automated Database Backups):**
  - **Frequency:** The system shall perform automated, full backups of the PostgreSQL database on a **daily** basis.
  - **Retention:** Daily backups shall be retained for a rolling period of at least **30 days**.
  - **Technology:** This will be managed using the automated snapshot feature of **Amazon RDS**.
- **NFR-BCK-002 (Automated File/Object Storage Backups):**
  - **Scope:** All user-uploaded files and digital assets stored in Amazon S3 (e.g., manuscript files, PDFs, images) shall be protected.
  - **Technology:** This will be achieved by enabling **versioning** on the S3 bucket and configuring cross-region replication for critical data. This ensures that files are protected against accidental deletion or modification.
- **NFR-BCK-003 (Backup Storage and Security):**
  - **Location:** All backups shall be stored in a geographically separate AWS region from the primary production environment to protect against regional disasters.
  - **Security:** All backup data, both at rest and in transit, must be encrypted.
- **NFR-BCK-004 (Backup Verification):**
  - **Testing:** The backup and restoration process shall be tested on a **quarterly** basis to ensure the integrity and reliability of the backups.

10.2 Disaster Recovery (DR) Plan

A formal Disaster Recovery (DR) plan will be in place to restore service in the event of a catastrophic failure of the primary production environment. The plan will adhere to the following objectives:

- **NFR-DR-001 (Recovery Time Objective - RTO):** The **RTO** for the platform shall be **24 hours**. This is the maximum acceptable time from the declaration of a disaster to the point when the ORA Digital Platform is fully operational again.
- **NFR-DR-002 (Recovery Point Objective - RPO):** The **RPO** for the platform shall be **24 hours**. This is the maximum acceptable amount of data loss, measured in time. Adherence to the daily backup schedule will ensure this objective is met.
- **NFR-DR-003 (DR Procedure):** The DR plan will include step-by-step procedures for:
    1. Declaring a disaster and activating the plan.
    2. Provisioning a new set of infrastructure in the designated recovery region.
    3. Restoring the PostgreSQL database from the latest successful daily backup.
    4. Restoring the file system from the replicated S3 bucket.
    5. Reconfiguring DNS settings to point to the new environment.
    6. Performing post-recovery testing to verify system functionality.

10.3 Maintenance Plan

A clear maintenance policy shall be established to handle routine updates, security patching, and system upgrades with minimal disruption to users.

- **NFR-MNT-001 (Scheduled Maintenance Windows):**
    - **Timing:** All planned maintenance activities that may require system downtime or result in degraded performance shall be scheduled during low-traffic periods (e.g., late night hours or weekends, based on user analytics).
    - **Notification:** Users shall be notified of any scheduled maintenance at least **48 hours in advance**. A notification banner will be displayed prominently on the platform's landing page.
- **NFR-MNT-002 (Security Patching):**
    - **Policy:** Critical security patches for the operating system, database, and all application dependencies shall be applied within a defined, timely manner upon their release, following a "patch and test" protocol in a staging environment before being deployed to production.
- **NFR-MNT-003 (Application Updates):**

- **Deployment:** The system will utilize a CI/CD pipeline to allow for zero-downtime (or minimal-downtime) deployments for most application updates and bug fixes, ensuring a seamless experience for users.